



? Dogs walk with much quicker strides than do humans. Is this primarily because, compared to human legs, dogs' legs (i) are shorter; (ii) are less massive; (iii) have a higher ratio of muscle to fat; (iv) have paws rather than toes; or (v) more than one of these?

14 Periodic Motion

Many kinds of motion repeat themselves over and over: the vibration of a quartz crystal in a watch, the swinging pendulum of a grandfather clock, the sound vibrations produced by a clarinet or an organ pipe, and the back-and-forth motion of the pistons in a car engine. This kind of motion, called **periodic motion** or **oscillation**, is the subject of this chapter. Understanding periodic motion will be essential for our later study of waves, sound, alternating electric currents, and light.

An object that undergoes periodic motion always has a stable equilibrium position. When it is moved away from this position and released, a force or torque comes into play to pull it back toward equilibrium. But by the time it gets there, it has picked up some kinetic energy, so it overshoots, stopping somewhere on the other side, and is again pulled back toward equilibrium. Picture a ball rolling back and forth in a round bowl or a pendulum that swings back and forth past its straight-down position.

In this chapter we'll concentrate on two simple examples of systems that can undergo periodic motions: spring-mass systems and pendulums. We'll also study why oscillations often tend to die out with time and why some oscillations can build up to greater and greater displacements from equilibrium when periodically varying forces act.

14.1 DESCRIBING OSCILLATION

Figure 14.1 (next page) shows one of the simplest systems that can have periodic motion. An object with mass m rests on a frictionless horizontal guide system, such as a linear air track, so it can move along the x -axis only. The object is attached to a spring of negligible mass that can be either stretched or compressed. The left end of the spring is held fixed, and the right end is attached to the object. The spring force is the only horizontal force acting on the object; the vertical normal and gravitational forces always add to zero.

It's simplest to define our coordinate system so that the origin O is at the equilibrium position, where the spring is neither stretched nor compressed. Then x is the x -component of the **displacement** of the object from equilibrium and is also the change in the length of the spring. The spring exerts a force on the object with x -component F_x , and the x -component of acceleration is $a_x = F_x/m$.

LEARNING OUTCOMES

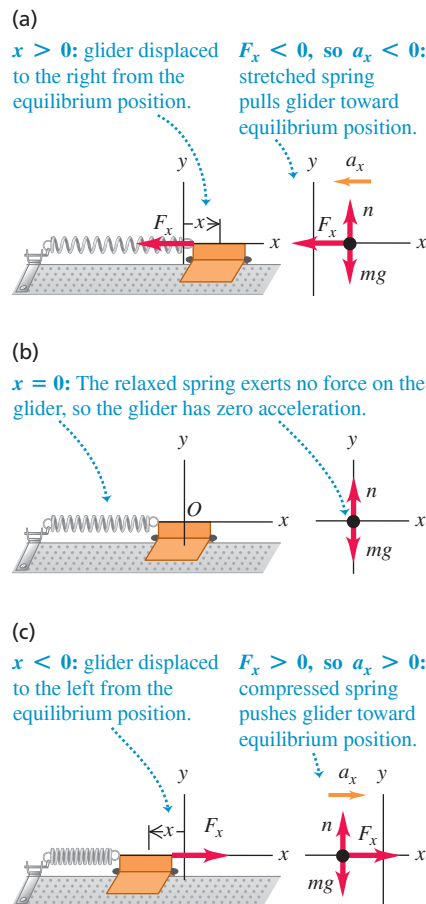
In this chapter, you'll learn...

- 14.1 How to describe oscillations in terms of amplitude, period, frequency, and angular frequency.
- 14.2 How to do calculations with simple harmonic motion, an important type of oscillation.
- 14.3 How to use energy concepts to analyze simple harmonic motion.
- 14.4 How to apply the ideas of simple harmonic motion to different physical situations.
- 14.5 How to analyze the motions of a simple pendulum.
- 14.6 What a physical pendulum is, and how to calculate the properties of its motion.
- 14.7 What determines how rapidly an oscillation dies out.
- 14.8 How a driving force applied to an oscillator at a particular frequency can cause a very large response, or resonance.

You'll need to review...

- 1.3 Time standards.
- 3.4 Uniform circular motion.
- 6.3 Hooke's law.
- 7.2, 7.4 Elastic potential energy; relating force and potential energy.
- 9.3 Relating angular motion and linear motion.
- 10.2 Newton's second law for rotational motion.

Figure 14.2 Model for periodic motion. When the object is displaced from its equilibrium position at $x = 0$, the spring exerts a restoring force back toward the equilibrium position.



BIO APPLICATION Wing Frequencies The ruby-throated hummingbird (*Archilochus colubris*) normally flaps its wings at about 50 Hz, producing the sound that gives hummingbirds their name. Insects can flap their wings at even faster rates, from 330 Hz for a house fly and 600 Hz for a mosquito to an amazing 1040 Hz for the tiny biting midge.

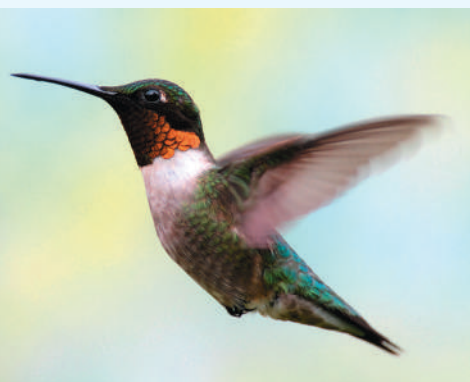


Figure 14.1 A system that can have periodic motion.

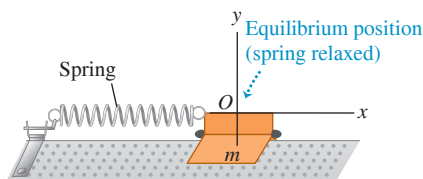


Figure 14.2 shows the object for three different displacements of the spring. Whenever the object is displaced from its equilibrium position, the spring force tends to restore it to the equilibrium position. We call a force with this character a **restoring force**. Oscillation can occur only when there is a restoring force tending to return the system to equilibrium.

Let's analyze how oscillation occurs in this system. If we displace the object to the right to $x = A$ and then let go, the net force and the acceleration are to the left (Fig. 14.2a). The speed increases as the object approaches the equilibrium position O . When the object is at O , the net force acting on it is zero (Fig. 14.2b), but because of its motion it *overshoots* the equilibrium position. On the other side of the equilibrium position the object is still moving to the left, but the net force and the acceleration are to the right (Fig. 14.2c); hence the speed decreases until the object comes to a stop. We'll show later that with an ideal spring, the stopping point is at $x = -A$. The object then accelerates to the right, overshoots equilibrium again, and stops at the starting point $x = A$, ready to repeat the whole process. The object is oscillating! If there is no friction or other force to remove mechanical energy from the system, this motion repeats forever; the restoring force perpetually draws the object back toward the equilibrium position, only to have the object overshoot time after time.

In different situations the force may depend on the displacement x from equilibrium in different ways. But oscillation *always* occurs if the force is a *restoring* force that tends to return the system to equilibrium.

Amplitude, Period, Frequency, and Angular Frequency

Here are some terms that we'll use in discussing periodic motions of all kinds:

The **amplitude** of the motion, denoted by A , is the maximum magnitude of displacement from equilibrium—that is, the maximum value of $|x|$. It is always positive. If the spring in Fig. 14.2 is an ideal one, the total overall range of the motion is $2A$. The SI unit of A is the meter. A complete vibration, or **cycle**, is one complete round trip—say, from A to $-A$ and back to A , or from O to A , back through O to $-A$, and back to O . Note that motion from one side to the other (say, $-A$ to A) is a half-cycle, not a whole cycle.

The **period**, T , is the time to complete one cycle. It is always positive. The SI unit is the second, but it is sometimes expressed as “seconds per cycle.”

The **frequency**, f , is the number of cycles in a unit of time. It is always positive. The SI unit of frequency is the *hertz*, named for the 19th-century German physicist Heinrich Hertz:

$$1 \text{ hertz} = 1 \text{ Hz} = 1 \text{ cycle/s} = 1 \text{ s}^{-1}$$

The **angular frequency**, ω , is 2π times the frequency:

$$\omega = 2\pi f$$

We'll learn shortly why ω is a useful quantity. It represents the rate of change of an angular quantity (not necessarily related to a rotational motion) that is always measured in radians, so its units are rad/s. Since f is in cycle/s, we may regard the number 2π as having units rad/cycle.

By definition, period and frequency are reciprocals of each other:

$$\text{In periodic motion frequency and period are reciprocals of each other.} \quad f = \frac{1}{T} \quad T = \frac{1}{f} \quad (14.1)$$

Also, from the definition of ω ,

$$\omega = 2\pi f = \frac{2\pi}{T} \quad (14.2)$$

Angular frequency related to frequency and period

Frequency

Period

CAUTION One period spans a complete cycle Keep in mind that the period of an oscillation is the time for a complete cycle—for example, the time to travel from $x = -A$ to $x = +A$ and back again to $x = -A$.

EXAMPLE 14.1 Period, frequency, and angular frequency

An ultrasonic transducer used for medical diagnosis oscillates at $6.7 \text{ MHz} = 6.7 \times 10^6 \text{ Hz}$. How long does each oscillation take, and what is the angular frequency?

IDENTIFY and SET UP The target variables are the period T and the angular frequency ω . We can find these from the given frequency f in Eqs. (14.1) and (14.2).

EXECUTE From Eqs. (14.1) and (14.2),

$$T = \frac{1}{f} = \frac{1}{6.7 \times 10^6 \text{ Hz}} = 1.5 \times 10^{-7} \text{ s} = 0.15 \mu\text{s}$$

$$\omega = 2\pi f = 2\pi(6.7 \times 10^6 \text{ Hz})$$

$$= (2\pi \text{ rad/cycle})(6.7 \times 10^6 \text{ cycle/s}) = 4.2 \times 10^7 \text{ rad/s}$$

EVALUATE This is a very rapid vibration, with large f and ω and small T . A slow vibration has small f and ω and large T .

KEYCONCEPT The period of an oscillation is the reciprocal of the oscillation frequency. The angular frequency equals the frequency multiplied by 2π .

TEST YOUR UNDERSTANDING OF SECTION 14.1 An object like that shown in Fig. 14.2 oscillates back and forth. For each of the following values of the object's x -velocity v_x and x -acceleration a_x , state whether its displacement x is positive, negative, or zero. (a) $v_x > 0$ and $a_x > 0$; (b) $v_x > 0$ and $a_x < 0$; (c) $v_x < 0$ and $a_x > 0$; (d) $v_x < 0$ and $a_x < 0$; (e) $v_x = 0$ and $a_x < 0$; (f) $v_x > 0$ and $a_x = 0$.

ANSWER

Figure 14.2 shows that (a) $v_x > 0$ and $a_x > 0$ (b) $v_x > 0$ and $a_x < 0$ (c) $v_x < 0$ and $a_x > 0$ (d) $v_x < 0$ and $a_x < 0$ (e) $v_x = 0$ and $a_x < 0$ (f) $v_x > 0$ and $a_x = 0$. The net force must also be zero and so the spring must be relaxed; hence $x = 0$.

14.2 SIMPLE HARMONIC MOTION

The simplest kind of oscillation occurs when the restoring force F_x is *directly proportional* to the displacement from equilibrium x . This happens if the spring in Figs. 14.1 and 14.2 is an ideal one that obeys *Hooke's law* (see Section 6.3). The constant of proportionality between F_x and x is the force constant k . On either side of the equilibrium position, F_x and x always have opposite signs. In Section 6.3 we represented the force acting *on* a stretched ideal spring as $F_x = kx$. The x -component of force the spring exerts *on the object* is the negative of this, so

$$F_x = -kx \quad (14.3)$$

Restoring force exerted by an ideal spring

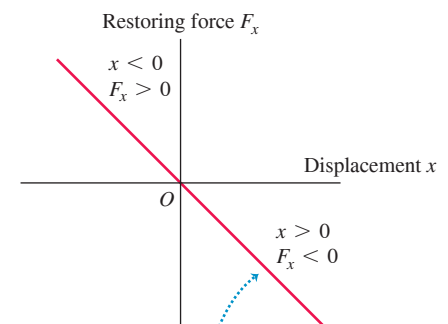
x -component of force

Displacement

Force constant of spring

This equation gives the correct magnitude and sign of the force, whether x is positive, negative, or zero (**Fig. 14.3**). The force constant k is always positive and has units of N/m (a useful alternative set of units is kg/s^2). We are assuming that there is no friction, so Eq. (14.3) gives the *net* force on the object.

Figure 14.3 An idealized spring exerts a restoring force that obeys Hooke's law, $F_x = -kx$. Oscillation with such a restoring force is called simple harmonic motion.



The restoring force exerted by an idealized spring is directly proportional to the displacement (Hooke's law, $F_x = -kx$): the graph of F_x versus x is a straight line.

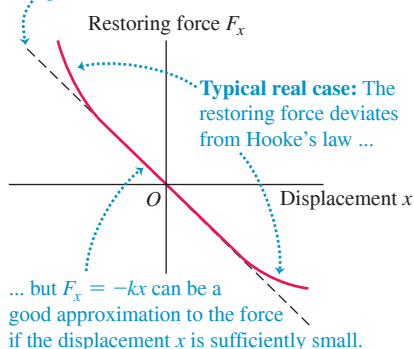
When the restoring force is directly proportional to the displacement from equilibrium, as given by Eq. (14.3), the oscillation is called **simple harmonic motion (SHM)**. The acceleration $a_x = d^2x/dt^2 = F_x/m$ of an object in SHM is

$$\text{Equation for simple harmonic motion} \quad a_x = \frac{d^2x}{dt^2} = -\frac{k}{m}x \quad (14.4)$$

a_x : x-component of acceleration
 $\frac{d^2x}{dt^2}$: Second derivative of displacement
 k : Force constant of restoring force
 x : Displacement
 m : Mass of object

Figure 14.4 In most real oscillations Hooke's law applies provided the object doesn't move too far from equilibrium. In such a case small-amplitude oscillations are approximately simple harmonic.

Ideal case: The restoring force obeys Hooke's law ($F_x = -kx$), so the graph of F_x versus x is a straight line.



The minus sign means that, in SHM, the acceleration and displacement always have opposite signs. This acceleration is *not* constant, so don't even think of using the constant-acceleration equations from Chapter 2. We'll see shortly how to solve this equation to find the displacement x as a function of time. An object that undergoes simple harmonic motion is called a **harmonic oscillator**.

Why is simple harmonic motion important? Not all periodic motions are simple harmonic; in periodic motion in general, the restoring force depends on displacement in a more complicated way than in Eq. (14.3). But in many systems the restoring force is *approximately* proportional to displacement if the displacement is sufficiently small (**Fig. 14.4**). That is, if the amplitude is small enough, the oscillations of such systems are approximately simple harmonic and therefore approximately described by Eq. (14.4). Thus we can use SHM as an approximate model for many different periodic motions, such as the vibration of a tuning fork, the electric current in an alternating-current circuit, and the oscillations of atoms in molecules and solids.

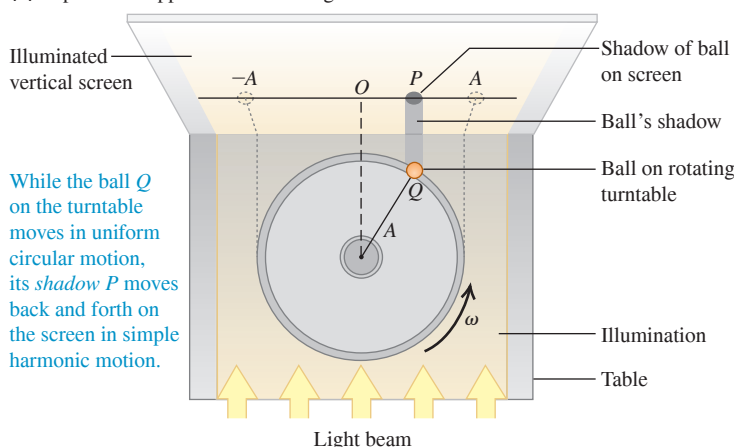
Circular Motion and the Equations of SHM

To explore the properties of simple harmonic motion, we must express the displacement x of the oscillating object as a function of time, $x(t)$. The second derivative of this function, d^2x/dt^2 , must be equal to $(-k/m)$ times the function itself, as required by Eq. (14.4). As we mentioned, the formulas for constant acceleration from Section 2.4 are no help because the acceleration changes constantly as the displacement x changes. Instead, we'll find $x(t)$ by noting that SHM is related to *uniform circular motion*, which we studied in Section 3.4.

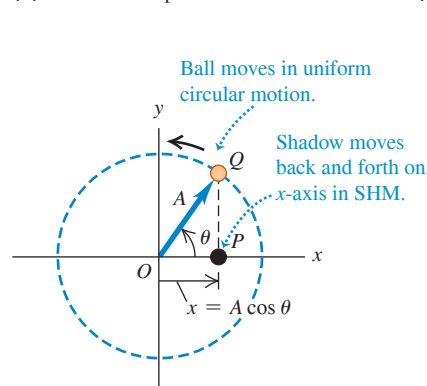
Figure 14.5a shows a top view of a horizontal disk of radius A with a ball attached to its rim at point Q . The disk rotates with constant angular speed ω (measured in rad/s), so the ball moves in uniform circular motion. A horizontal light beam casts a shadow of the ball on a screen. The shadow at point P oscillates back and forth as the ball moves in a circle. We then arrange an object attached to an ideal spring, like the combination shown in

Figure 14.5 (a) Relating uniform circular motion and simple harmonic motion. (b) The ball's shadow moves exactly like an object oscillating on an ideal spring.

(a) Top view of apparatus for creating the reference circle



(b) An abstract representation of the motion in (a)



Figs. 14.1 and 14.2, so that the object oscillates parallel to the shadow. We'll prove that the motions of the object and of the ball's shadow are *identical* if the amplitude of the object's oscillation is equal to the disk radius A , and if the angular frequency $2\pi f$ of the oscillating object is equal to the angular speed ω of the rotating disk. That is, *simple harmonic motion is the projection of uniform circular motion onto a diameter*.

We can verify this remarkable statement by finding the acceleration of the shadow at P and comparing it to the acceleration of an object undergoing SHM, given by Eq. (14.4). The circle in which the ball moves so that its projection matches the motion of the oscillating object is called the **reference circle**; we'll call the point Q the *reference point*. We take the reference circle to lie in the xy -plane, with the origin O at the center of the circle (Fig. 14.5b). At time t the vector OQ from the origin to reference point Q makes an angle θ with the positive x -axis. As point Q moves around the reference circle with constant angular speed ω , vector OQ rotates with the same angular speed. Such a rotating vector is called a **phasor**. (This term was in use long before the invention of the *Star Trek* stun gun with a similar name.) We'll use phasors again when we study alternating-current circuits in Chapter 31 and the interference of light in Chapters 35 and 36.

The x -component of the phasor at time t is just the x -coordinate of the point Q :

$$x = A \cos \theta \quad (14.5)$$

This is also the x -coordinate of the shadow P , which is the *projection* of Q onto the x -axis. Hence the x -velocity of the shadow P along the x -axis is equal to the x -component of the velocity vector of point Q (Fig. 14.6a), and the x -acceleration of P is equal to the x -component of the acceleration vector of Q (Fig. 14.6b). Since point Q is in uniform circular motion, its acceleration vector $\vec{a}_Q$ is always directed toward O . Furthermore, the magnitude of $\vec{a}_Q$ is constant and given by the angular speed squared times the radius of the circle (see Section 9.3):

$$a_Q = \omega^2 A \quad (14.6)$$

Figure 14.6b shows that the x -component of $\vec{a}_Q$ is $a_x = -a_Q \cos \theta$. Combining this with Eqs. (14.5) and (14.6), we get that the acceleration of point P is

$$a_x = -a_Q \cos \theta = -\omega^2 A \cos \theta \quad \text{or} \quad (14.7)$$

$$a_x = -\omega^2 x \quad (14.8)$$

The acceleration of point P is directly proportional to the displacement x and always has the opposite sign. These are precisely the hallmarks of simple harmonic motion.

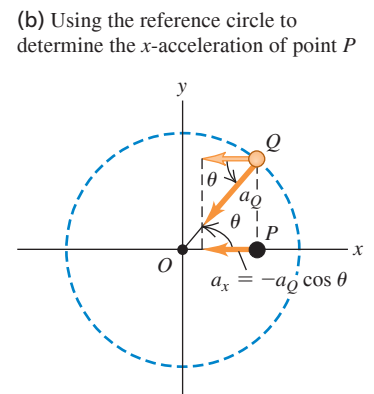
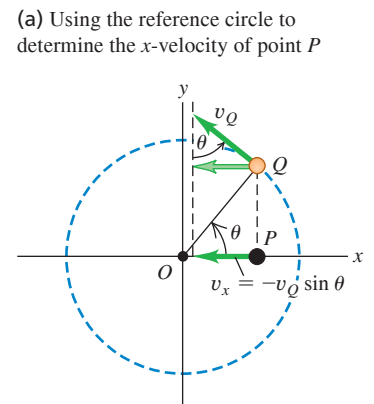
Equation (14.8) is *exactly* the same as Eq. (14.4) for the acceleration of a harmonic oscillator, provided that the angular speed ω of the reference point Q is related to the force constant k and mass m of the oscillating object by

$$\omega^2 = \frac{k}{m} \quad \text{or} \quad \omega = \sqrt{\frac{k}{m}} \quad (14.9)$$

We have been using the same symbol ω for the angular *speed* of the reference point Q and the angular *frequency* of the oscillating point P . The reason is that these quantities are equal! If point Q makes one complete revolution in time T , then point P goes through one complete cycle of oscillation in the same time; hence T is the period of the oscillation. During time T the point Q moves through 2π radians, so its angular speed is $\omega = 2\pi/T$. But this is the same as Eq. (14.2) for the angular frequency of the point P , which verifies our statement about the two interpretations of ω . This is why we introduced angular frequency in Section 14.1; this quantity makes the connection between oscillation and circular motion. So we reinterpret Eq. (14.9) as an expression for the angular frequency of simple harmonic motion:

$$\text{Angular frequency for simple harmonic motion } \omega = \sqrt{\frac{k}{m}} \quad \begin{array}{l} \leftarrow \text{Force constant of restoring force} \\ \leftarrow \text{Mass of object} \end{array} \quad (14.10)$$

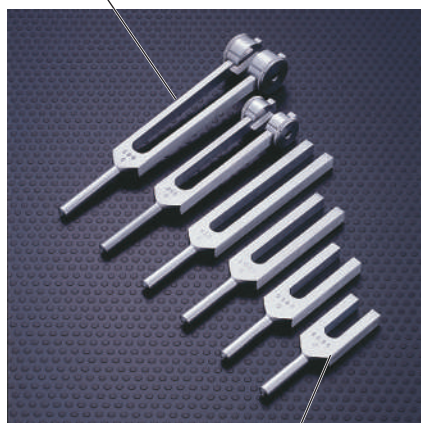
Figure 14.6 The (a) x -velocity and (b) x -acceleration of the ball's shadow P (see Fig. 14.5) are the x -components of the velocity and acceleration vectors, respectively, of the ball Q .



CAUTION Don't confuse frequency and angular frequency You can run into trouble if you don't make the distinction between frequency f and angular frequency $\omega = 2\pi f$. Frequency tells you how many cycles of oscillation occur per second, while angular frequency tells you how many radians per second this corresponds to on the reference circle. In solving problems, pay careful attention to whether the goal is to find f or ω .

Figure 14.7 The greater the mass m in a tuning fork's tines, the lower the frequency of oscillation $f = (1/2\pi)\sqrt{k/m}$ and the lower the pitch of the sound that the tuning fork produces.

Tines with large mass m :
low frequency $f = 128$ Hz



Tines with small mass m :
high frequency $f = 4096$ Hz

When you start an object oscillating in SHM, the value of ω is not yours to choose; it is predetermined by the values of k and m . The units of k are N/m or kg/s², so k/m is in (kg/s²)/kg = s⁻². When we take the square root in Eq. (14.10), we get s⁻¹, or more properly rad/s because this is an *angular* frequency (recall that a radian is not a true unit).

According to Eqs. (14.1) and (14.2), the frequency f and period T are

$$\text{Frequency for simple harmonic motion } f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (14.11)$$

Angular frequency Force constant of restoring force Mass of object

$$\text{Period for simple harmonic motion } T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad (14.12)$$

Mass of object Force constant of restoring force Frequency Angular frequency

We see from Eq. (14.12) that a larger mass m will have less acceleration and take a longer time for a complete cycle (Fig. 14.7). A stiffer spring (one with a larger force constant k) exerts a greater force at a given deformation x , causing greater acceleration and a shorter time T per cycle.

Period and Amplitude in SHM

Equations (14.11) and (14.12) show that the period and frequency of simple harmonic motion are completely determined by the mass m and the force constant k . *In simple harmonic motion the period and frequency do not depend on the amplitude A .* For given values of m and k , the time of one complete oscillation is the same whether the amplitude is large or small. Equation (14.3) shows why we should expect this. Larger A means that the object reaches larger values of $|x|$ and is subjected to larger restoring forces. This increases the average speed of the object over a complete cycle; this exactly compensates for having to travel a larger distance, so the same total time is involved.

The oscillations of a tuning fork are essentially simple harmonic motion, so it always vibrates with the same frequency, independent of amplitude. This is why a tuning fork can be used as a standard for musical pitch. If it were not for this characteristic of simple harmonic motion, it would be impossible to play most musical instruments in tune. If you encounter an oscillating object with a period that *does* depend on the amplitude, the oscillation is *not* simple harmonic motion.

EXAMPLE 14.2 Angular frequency, frequency, and period in SHM

WITH VARIATION PROBLEMS

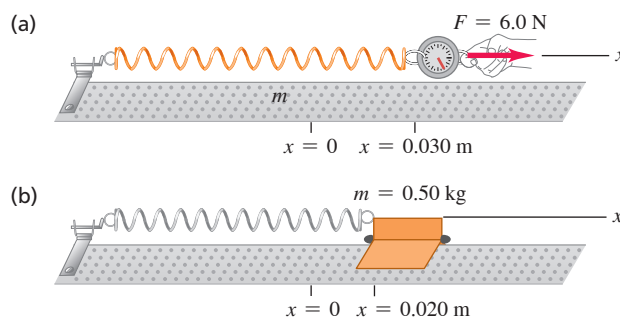
A spring is mounted horizontally, with its left end fixed. A spring balance attached to the free end and pulled toward the right (Fig. 14.8a) indicates that the stretching force is proportional to the displacement, and a force of 6.0 N causes a displacement of 0.030 m. We replace the spring balance with a 0.50 kg glider, pull it 0.020 m to the right along a frictionless air track, and release it from rest (Fig. 14.8b). (a) Find the force constant k of the spring. (b) Find the angular frequency ω , frequency f , and period T of the resulting oscillation.

IDENTIFY and SET UP Because the spring force (equal in magnitude to the stretching force) is proportional to the displacement, the motion is simple harmonic. We find k from Hooke's law, Eq. (14.3), and ω , f , and T from Eqs. (14.10), (14.11), and (14.12), respectively.

EXECUTE (a) When $x = 0.030$ m, the force the spring exerts on the spring balance is $F_x = -6.0$ N. From Eq. (14.3),

$$k = -\frac{F_x}{x} = -\frac{-6.0 \text{ N}}{0.030 \text{ m}} = 200 \text{ N/m} = 200 \text{ kg/s}^2$$

Figure 14.8 (a) The force exerted on the spring (shown by the vector F) has x -component $F_x = +6.0$ N. The force exerted by the spring has x -component $F_x = -6.0$ N. (b) A glider is attached to the same spring and allowed to oscillate.



(b) From Eqs. (14.10), (14.11), and (14.12), with $m = 0.50 \text{ kg}$,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{200 \text{ kg/s}^2}{0.50 \text{ kg}}} = 20 \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{20 \text{ rad/s}}{2\pi \text{ rad/cycle}} = 3.2 \text{ cycle/s} = 3.2 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{3.2 \text{ cycle/s}} = 0.31 \text{ s}$$

EVALUATE The amplitude of the oscillation is 0.020 m , the distance that we pulled the glider before releasing it. In SHM the angular frequency, frequency, and period are all independent of the amplitude. Note that a period is usually stated in “seconds” rather than “seconds per cycle.”

KEYCONCEPT The magnitude of the force exerted by an ideal spring equals the spring constant k times the distance the spring is stretched or compressed. An object of mass m attached to an ideal spring oscillates in simple harmonic motion, with its period, frequency, and angular frequency determined by the values of k and m .

Displacement, Velocity, and Acceleration in SHM

We still need to find the displacement x as a function of time for a harmonic oscillator. Equation (14.4) for an object in SHM along the x -axis is identical to Eq. (14.8) for the x -coordinate of the reference point in uniform circular motion with constant angular speed $\omega = \sqrt{k/m}$. Hence Eq. (14.5), $x = A \cos \theta$, describes the x -coordinate for both situations. If at $t = 0$ the phasor OQ makes an angle ϕ (the Greek letter phi) with the positive x -axis, then at any later time t this angle is $\theta = \omega t + \phi$. We substitute this into Eq. (14.5) to obtain

$$x = A \cos(\omega t + \phi) \quad (14.13)$$

Displacement in simple harmonic motion as a function of time Amplitude Time Phase angle
Angular frequency $= \sqrt{k/m}$

Figure 14.9 shows a graph of Eq. (14.13) for the particular case $\phi = 0$. We could also have written Eq. (14.13) in terms of a sine function rather than a cosine by using the identity $\cos \alpha = \sin(\alpha + \pi/2)$. *In simple harmonic motion the displacement is a periodic, sinusoidal function of time.* There are many other periodic functions, but none so simple as a sine or cosine function.

The value of the cosine function is always between -1 and 1 , so in Eq. (14.13), x is always between $-A$ and A . This confirms that A is the amplitude of the motion.

The cosine function in Eq. (14.13) repeats itself whenever time t increases by one period T , or when $\omega t + \phi$ increases by 2π radians. Thus, if we start at time $t = 0$, the time T to complete one cycle is

$$\omega T = \sqrt{\frac{k}{m}} T = 2\pi \quad \text{or} \quad T = 2\pi \sqrt{\frac{m}{k}}$$

which is just Eq. (14.12). Changing either m or k changes the period T (**Figs. 14.10a** and **14.10b**), but T does not depend on the amplitude A (**Fig. 14.10c**).

Figure 14.9 Graph of x versus t [see Eq. (14.13)] for simple harmonic motion. The case shown has $\phi = 0$.

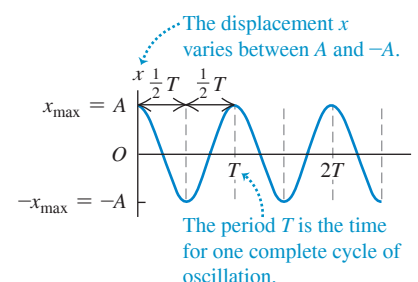
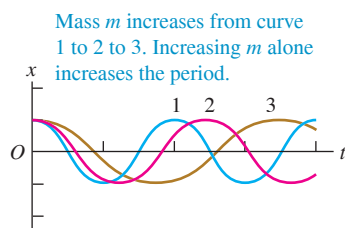
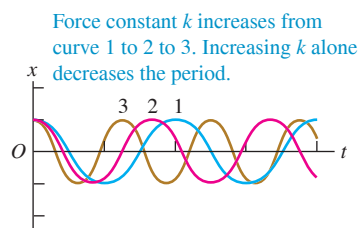


Figure 14.10 Variations of simple harmonic motion. All cases shown have $\phi = 0$ [see Eq. (14.13)].

(a) Increasing m ; same A and k



(b) Increasing k ; same A and m



(c) Increasing A ; same k and m

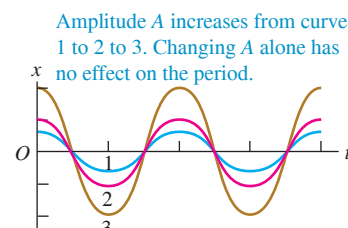
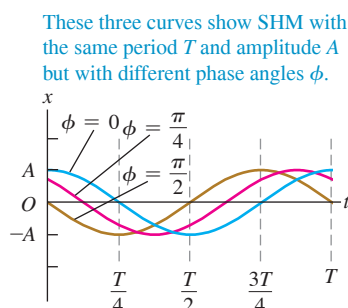


Figure 14.11 Variations of simple harmonic motion: same m , k , and A but different phase angles ϕ .



The constant ϕ in Eq. (14.13) is called the **phase angle**. It tells us at what point in the cycle the motion was at $t = 0$ (equivalent to where around the circle the point Q was at $t = 0$). We denote the displacement at $t = 0$ by x_0 . Putting $t = 0$ and $x = x_0$ in Eq. (14.13), we get

$$x_0 = A \cos \phi \quad (14.14)$$

If $\phi = 0$, then $x_0 = A \cos 0 = A$, and the object starts at its maximum positive displacement. If $\phi = \pi$, then $x_0 = A \cos \pi = -A$, and the particle starts at its maximum *negative* displacement. If $\phi = \pi/2$, then $x_0 = A \cos(\pi/2) = 0$, and the particle is initially at the origin. **Figure 14.11** shows the displacement x versus time for three different phase angles.

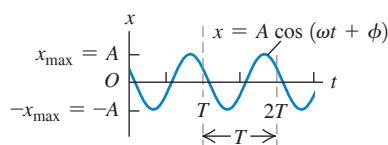
We find the velocity v_x and acceleration a_x as functions of time for a harmonic oscillator by taking derivatives of Eq. (14.13) with respect to time:

$$v_x = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi) \quad (\text{velocity in SHM}) \quad (14.15)$$

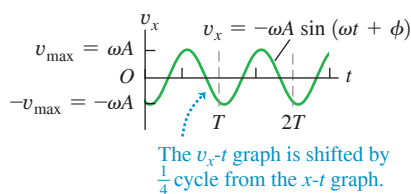
$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \phi) \quad (\text{acceleration in SHM}) \quad (14.16)$$

Figure 14.12 Graphs of (a) x versus t , (b) v_x versus t , and (c) a_x versus t for an object in SHM. For the motion depicted in these graphs, $\phi = \pi/3$.

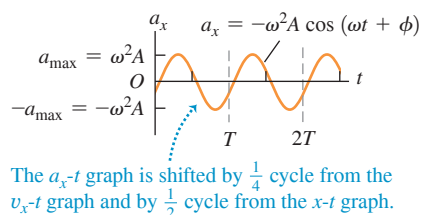
(a) Displacement x as a function of time t



(b) Velocity v_x as a function of time t



(c) Acceleration a_x as a function of time t



The velocity v_x oscillates between

$$v_{\max} = +\omega A \quad \text{and} \quad -v_{\max} = -\omega A,$$

and the acceleration a_x oscillates between

$$a_{\max} = +\omega^2 A \quad \text{and} \quad -a_{\max} = -\omega^2 A$$

(Fig. 14.12). Comparing Eq. (14.16) with Eq. (14.13) and recalling that $\omega^2 = k/m$ from Eq. (14.9), we see that

$$a_x = -\omega^2 x = -\frac{k}{m}x$$

which is just Eq. (14.4) for simple harmonic motion. This confirms that Eq. (14.13) for x as a function of time is correct.

We actually derived Eq. (14.16) earlier in a geometrical way by taking the x -component of the acceleration vector of the reference point Q . This was done in Fig. 14.6b and Eq. (14.7) (recall that $\theta = \omega t + \phi$). In the same way, we could have derived Eq. (14.15) by taking the x -component of the velocity vector of Q , as shown in Fig. 14.6b. We'll leave the details for you to work out.

Note that the sinusoidal graph of displacement versus time (Fig. 14.12a) is shifted by one-quarter period from the graph of velocity versus time (Fig. 14.12b) and by one-half

period from the graph of acceleration versus time (Fig. 14.12c). **Figure 14.13** shows why this is so. When the object is passing through the equilibrium position so that $x = 0$, the velocity equals either $v_{\max}$ or $-v_{\max}$ (depending on which way the object is moving) and the acceleration is zero. When the object is at either its most positive displacement, $x = +A$, or its most negative displacement, $x = -A$, the velocity is zero and the object is instantaneously at rest. At these points, the restoring force $F_x = -kx$ and the acceleration of the object have their maximum magnitudes. At $x = +A$ the acceleration is negative and equal to $-a_{\max}$. At $x = -A$ the acceleration is positive: $a_x = +a_{\max}$.

Here's how we can determine the amplitude A and phase angle ϕ for an oscillating object if we are given its initial displacement x_0 and initial velocity v_{0x} . The initial velocity v_{0x} is the velocity at time $t = 0$; putting $v_x = v_{0x}$ and $t = 0$ in Eq. (14.15), we find

$$v_{0x} = -\omega A \sin \phi \quad (14.17)$$

To find ϕ , we divide Eq. (14.17) by Eq. (14.14). This eliminates A and gives an equation that we can solve for ϕ :

$$\frac{v_{0x}}{x_0} = \frac{-\omega A \sin \phi}{A \cos \phi} = -\omega \tan \phi$$

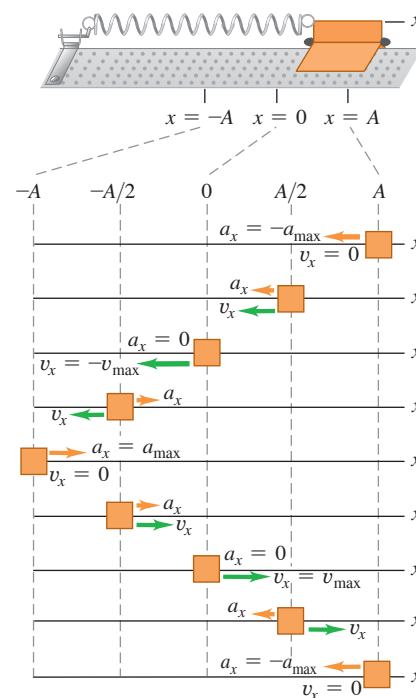
$$\phi = \arctan\left(-\frac{v_{0x}}{\omega x_0}\right) \quad (\text{phase angle in SHM}) \quad (14.18)$$

It is also easy to find the amplitude A if we are given x_0 and v_{0x} . We'll sketch the derivation, and you can fill in the details. Square Eq. (14.14); then divide Eq. (14.17) by ω , square it, and add to the square of Eq. (14.14). The right side will be $A^2(\sin^2 \phi + \cos^2 \phi)$, which is equal to A^2 . The final result is

$$A = \sqrt{x_0^2 + \frac{v_{0x}^2}{\omega^2}} \quad (\text{amplitude in SHM}) \quad (14.19)$$

Note that when the object has both an initial displacement x_0 and a nonzero initial velocity v_{0x} , the amplitude A is *not* equal to the initial displacement. That's reasonable; if you start the object at a positive x_0 but give it a positive velocity v_{0x} , it will go *farther* than x_0 before it turns and comes back, and so $A > x_0$.

Figure 14.13 How x -velocity v_x and x -acceleration a_x vary during one cycle of SHM.



PROBLEM-SOLVING STRATEGY 14.1 Simple Harmonic Motion I: Describing Motion

IDENTIFY the relevant concepts: An oscillating system undergoes simple harmonic motion (SHM) *only* if the restoring force is directly proportional to the displacement.

SET UP the problem using the following steps:

1. Identify the known and unknown quantities, and determine which are the target variables.
2. Distinguish between two kinds of quantities. *Properties of the system* include the mass m , the force constant k , and quantities derived from m and k , such as the period T , frequency f , and angular frequency ω . These are independent of *properties of the motion*, which describe how the system behaves when it is set into motion in a particular way; they include the amplitude A , maximum velocity $v_{\max}$, and phase angle ϕ , and values of x , v_x , and a_x at particular times.
3. If necessary, define an x -axis as in Fig. 14.13, with the equilibrium position at $x = 0$.

EXECUTE the solution as follows:

1. Use the equations given in Sections 14.1 and 14.2 to solve for the target variables.
2. To find the values of x , v_x , and a_x at particular times, use Eqs. (14.13), (14.15), and (14.16), respectively. If both the initial displacement x_0 and initial velocity v_{0x} are given, determine ϕ and A from Eqs. (14.18) and (14.19). If the object has an initial positive displacement x_0 but zero initial velocity ($v_{0x} = 0$), then the amplitude is $A = x_0$ and the phase angle is $\phi = 0$. If it has an initial positive velocity v_{0x} but no initial displacement ($x_0 = 0$), the amplitude is $A = v_{0x}/\omega$ and the phase angle is $\phi = -\pi/2$. Express all phase angles in *radians*.

EVALUATE your answer: Make sure that your results are consistent. For example, suppose you used x_0 and v_{0x} to find general expressions for x and v_x at time t . If you substitute $t = 0$ into these expressions, you should get back the given values of x_0 and v_{0x} .

EXAMPLE 14.3 Describing SHM

WITH VARIATION PROBLEMS

We give the glider of Example 14.2 an initial displacement $x_0 = +0.015$ m and an initial velocity $v_{0x} = +0.40$ m/s. (a) Find the period, amplitude, and phase angle of the resulting motion. (b) Write equations for the displacement, velocity, and acceleration as functions of time.

IDENTIFY and SET UP As in Example 14.2, the oscillations are SHM. We use equations from this section and the given values $k = 200$ N/m, $m = 0.50$ kg, x_0 , and v_{0x} to calculate the target variables A and ϕ and to obtain expressions for x , v_x , and a_x .

EXECUTE (a) In SHM the period and angular frequency are *properties of the system* that depend on only k and m , not on the amplitude, and so are the same as in Example 14.2 ($T = 0.31$ s and $\omega = 20$ rad/s). From Eq. (14.19), the amplitude is

$$A = \sqrt{x_0^2 + \frac{v_{0x}^2}{\omega^2}} = \sqrt{(0.015 \text{ m})^2 + \frac{(0.40 \text{ m/s})^2}{(20 \text{ rad/s})^2}} = 0.025 \text{ m}$$

We use Eq. (14.18) to find the phase angle:

$$\begin{aligned}\phi &= \arctan\left(-\frac{v_{0x}}{\omega x_0}\right) \\ &= \arctan\left(-\frac{0.40 \text{ m/s}}{(20 \text{ rad/s})(0.015 \text{ m})}\right) = -53^\circ = -0.93 \text{ rad}\end{aligned}$$

(b) The displacement, velocity, and acceleration at any time are given by Eqs. (14.13), (14.15), and (14.16), respectively. We substitute the values of A , ω , and ϕ into these equations:

$$\begin{aligned}x &= (0.025 \text{ m}) \cos[(20 \text{ rad/s})t - 0.93 \text{ rad}] \\ v_x &= -(0.50 \text{ m/s}) \sin[(20 \text{ rad/s})t - 0.93 \text{ rad}] \\ a_x &= -(10 \text{ m/s}^2) \cos[(20 \text{ rad/s})t - 0.93 \text{ rad}]\end{aligned}$$

EVALUATE You can check the expressions for x and v_x by confirming that if you substitute $t = 0$, they yield $x = x_0 = 0.015$ m and $v_x = v_{0x} = 0.40$ m/s.

KEYCONCEPT You can determine the amplitude and phase angle of a simple harmonic oscillation from the initial position, initial velocity, and angular frequency of the motion.

TEST YOUR UNDERSTANDING OF SECTION 14.2 A glider is attached to a spring as shown in Fig. 14.13. If the glider is moved to $x = 0.10$ m and released from rest at time $t = 0$, it will oscillate with amplitude $A = 0.10$ m and phase angle $\phi = 0$. (a) Suppose instead that at $t = 0$ the glider is at $x = 0.10$ m and is moving to the right in Fig. 14.13. In this situation is the amplitude greater than, less than, or equal to 0.10 m? Is the phase angle greater than, less than, or equal to zero? (b) Suppose instead that at $t = 0$ the glider is at $x = 0.10$ m and is moving to the left in Fig. 14.13. In this situation is the amplitude greater than, less than, or equal to 0.10 m? Is the phase angle greater than, less than, or equal to zero?

ANSWER (a) $0 < \phi < \pi$ and $0 < A < 0.10$ m. (b) $0 < \phi < \pi$ and $0 < A < 0.10$ m. In part (a), the glider is moving to the right at $t = 0$, so the initial velocity v_{0x} is positive. The initial displacement x_0 is also positive. From Eq. (14.18), the phase angle ϕ is positive. In part (b), the glider is moving to the left at $t = 0$, so the initial velocity v_{0x} is negative. The initial displacement x_0 is also positive. From Eq. (14.18), the phase angle ϕ is positive. In both situations the initial velocity v_{0x} is nonzero, so the amplitude A is greater than 0.10 m. In both situations the initial displacement x_0 is positive, so the amplitude A is greater than 0.10 m.

14.3 ENERGY IN SIMPLE HARMONIC MOTION

We can learn even more about simple harmonic motion by using energy considerations. The only horizontal force on the object in SHM in Figs. 14.2 and 14.13 is the conservative force exerted by an ideal spring. The vertical forces do no work, so the total mechanical energy of the system is *conserved*. We also assume that the mass of the spring itself is negligible.

The kinetic energy of the object is $K = \frac{1}{2}mv^2$ and the potential energy of the spring is $U = \frac{1}{2}kx^2$, just as in Section 7.2. There are no nonconservative forces that do work, so the total mechanical energy $E = K + U$ is conserved:

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \text{constant} \quad (14.20)$$

(Since the motion is one-dimensional, $v^2 = v_x^2$.)

The total mechanical energy E is also directly related to the amplitude A of the motion. When the object reaches the point $x = A$, its maximum displacement from equilibrium, it momentarily stops as it turns back toward the equilibrium position. That is, when $x = A$ (or $-A$), $v_x = 0$. At this point the energy is entirely potential, and $E = \frac{1}{2}kA^2$. Because E is constant, it is equal to $\frac{1}{2}kA^2$ at any other point. Combining this expression with Eq. (14.20), we get

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \text{constant} \quad (14.21)$$

Mass
Force constant of restoring force
Velocity
Displacement
Amplitude

We can verify this equation by substituting x and v_x from Eqs. (14.13) and (14.15) and using $\omega^2 = k/m$ from Eq. (14.9):

$$\begin{aligned} E &= \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}m[-\omega A \sin(\omega t + \phi)]^2 + \frac{1}{2}k[A \cos(\omega t + \phi)]^2 \\ &= \frac{1}{2}kA^2 \sin^2(\omega t + \phi) + \frac{1}{2}kA^2 \cos^2(\omega t + \phi) = \frac{1}{2}kA^2 \end{aligned}$$

(Recall that $\sin^2 \alpha + \cos^2 \alpha = 1$.) Hence our expressions for displacement and velocity in SHM are consistent with energy conservation, as they must be.

We can use Eq. (14.21) to solve for the velocity v_x of the object at a given displacement x :

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2} \quad (14.22)$$

The $\pm$ sign means that at a given value of x the object can be moving in either direction. For example, when $x = \pm A/2$,

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - \left(\pm \frac{A}{2}\right)^2} = \pm \sqrt{\frac{3}{4}} \sqrt{\frac{k}{m}} A$$

Equation (14.22) also shows that the *maximum* speed $v_{\max}$ occurs at $x = 0$. Using Eq. (14.10), $\omega = \sqrt{k/m}$, we find that

$$v_{\max} = \sqrt{\frac{k}{m}} A = \omega A \quad (14.23)$$

This agrees with Eq. (14.15): v_x oscillates between $-\omega A$ and $+\omega A$.

Interpreting E , K , and U in SHM

Figure 14.14 shows the energy quantities E , K , and U at $x = 0$, $x = \pm A/2$, and $x = \pm A$. **Figure 14.15** (next page) is a graphical display of Eq. (14.21); energy (kinetic, potential, and total) is plotted vertically and the coordinate x is plotted horizontally. The parabolic curve in Fig. 14.15a represents the potential energy $U = \frac{1}{2}kx^2$. The horizontal line represents the total mechanical energy E , which is constant and does not vary with x . At any value of x between $-A$ and A , the vertical distance from the x -axis to the parabola is U ; since $E = K + U$, the remaining vertical distance up to the horizontal line is K . Figure 14.15b shows both K and U as functions of x . The horizontal line for E intersects the potential-energy curve at $x = -A$ and $x = A$, so at these points the energy is entirely potential, the kinetic energy is zero, and the object comes momentarily to rest before reversing direction. As the object oscillates between $-A$ and A , the energy is continuously transformed from potential to kinetic and back again.

Figure 14.14 Graphs of E , K , and U versus displacement in SHM. The velocity of the object is *not* constant, so these images of the object at equally spaced positions are *not* equally spaced in time.

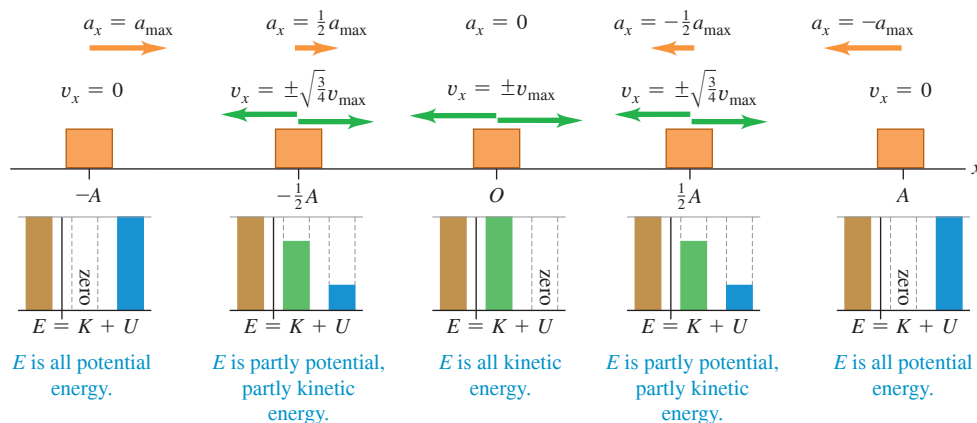
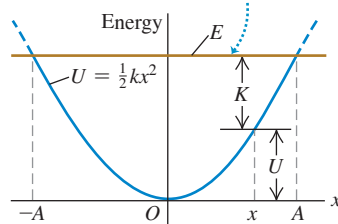


Figure 14.15 Kinetic energy K , potential energy U , and total mechanical energy E as functions of displacement for SHM. At each value of x the sum of the values of K and U equals the constant value of E . Can you show that the energy is half kinetic and half potential at $x = \pm\sqrt{\frac{1}{2}}A$?

(a) The potential energy U and total mechanical energy E for an object in SHM as a function of displacement x

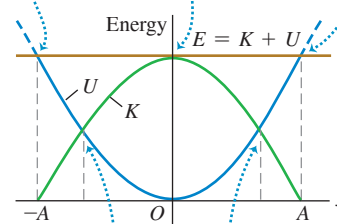
The total mechanical energy E is constant.



(b) The same graph as in (a), showing kinetic energy K as well

At $x = \pm A$ the energy is all potential; $K = 0$.

At $x = 0$ the energy is all kinetic; $U = 0$.



At these points the energy is half kinetic and half potential.

Figure 14.15a shows the connection between the amplitude A and the corresponding total mechanical energy $E = \frac{1}{2}kA^2$. If we tried to make x greater than A (or less than $-A$), U would be greater than E , and K would have to be negative. But K can never be negative, so x can't be greater than A or less than $-A$.

PROBLEM-SOLVING STRATEGY 14.2 Simple Harmonic Motion II: Energy

The SHM energy equation, Eq. (14.21), is a useful relationship among velocity, displacement, and total mechanical energy. If a problem requires you to relate displacement, velocity, and acceleration without reference to time, consider using Eq. (14.4) (from Newton's second law) or Eq. (14.21) (from energy conservation). Because Eq. (14.21)

involves x^2 and v_x^2 , you must infer the *signs* of x and v_x from the situation. For instance, if the object is moving from the equilibrium position toward the point of greatest positive displacement, then x is positive and v_x is positive.

EXAMPLE 14.4 Velocity, acceleration, and energy in SHM

WITH VARIATION PROBLEMS

(a) Find the maximum and minimum velocities attained by the oscillating glider of Example 14.2. (b) Find the maximum and minimum accelerations. (c) Find the velocity v_x and acceleration a_x when the glider is halfway from its initial position to the equilibrium position $x = 0$. (d) Find the total mechanical energy, potential energy, and kinetic energy at this position.

IDENTIFY and SET UP The problem concerns properties of the motion at specified *positions*, not at specified *times*, so we can use the energy relationships of this section. Figure 14.13 shows our choice of x -axis. The maximum displacement from equilibrium is $A = 0.020$ m. We use Eqs. (14.22) and (14.4) to find v_x and a_x for a given x . We then use Eq. (14.21) for given x and v_x to find the total, potential, and kinetic energies E , U , and K .

EXECUTE (a) From Eq. (14.22), the velocity v_x at any displacement x is

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2}$$

The glider's maximum *speed* occurs when it is moving through $x = 0$:

$$v_{\max} = \sqrt{\frac{k}{m}} A = \sqrt{\frac{200 \text{ N/m}}{0.50 \text{ kg}}} (0.020 \text{ m}) = 0.40 \text{ m/s}$$

Its maximum and minimum (most negative) *velocities* are $+0.40$ m/s and -0.40 m/s, which occur when it is moving through $x = 0$ to the right and left, respectively.

(b) From Eq. (14.4), $a_x = -(k/m)x$. The glider's maximum (most positive) acceleration occurs at the most negative value of x , $x = -A$:

$$a_{\max} = -\frac{k}{m}(-A) = -\frac{200 \text{ N/m}}{0.50 \text{ kg}}(-0.020 \text{ m}) = 8.0 \text{ m/s}^2$$

The minimum (most negative) acceleration is $a_{\min} = -8.0 \text{ m/s}^2$, which occurs at $x = +A = +0.020$ m.

(c) The point halfway from $x = x_0 = A$ to $x = 0$ is $x = A/2 = 0.010$ m. From Eq. (14.22), at this point

$$v_x = -\sqrt{\frac{200 \text{ N/m}}{0.50 \text{ kg}}} \sqrt{(0.020 \text{ m})^2 - (0.010 \text{ m})^2} = -0.35 \text{ m/s}$$

We choose the negative square root because the glider is moving from $x = A$ toward $x = 0$. From Eq. (14.4),

$$a_x = -\frac{200 \text{ N/m}}{0.50 \text{ kg}}(0.010 \text{ m}) = -4.0 \text{ m/s}^2$$

Figure 14.14 shows the conditions at $x = 0$, $\pm A/2$, and $\pm A$.

(d) The energies are

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(200 \text{ N/m})(0.020 \text{ m})^2 = 0.040 \text{ J}$$

$$U = \frac{1}{2}kx^2 = \frac{1}{2}(200 \text{ N/m})(0.010 \text{ m})^2 = 0.010 \text{ J}$$

$$K = \frac{1}{2}mv_x^2 = \frac{1}{2}(0.50 \text{ kg})(-0.35 \text{ m/s})^2 = 0.030 \text{ J}$$

EVALUATE At $x = A/2$, the total mechanical energy is one-fourth potential energy and three-fourths kinetic energy. You can confirm this by inspecting Fig. 14.15b.

KEYCONCEPT In simple harmonic motion, the sum of kinetic energy and elastic potential energy is conserved. You can use this to find the velocity of an object in SHM as a function of its displacement.

EXAMPLE 14.5 Energy and momentum in SHM

A block of mass M attached to a horizontal spring with force constant k is moving in SHM with amplitude A_1 . As the block passes through its equilibrium position, a lump of putty of mass m is dropped from a small height and sticks to it. (a) Find the new amplitude and period of the motion. (b) Repeat part (a) if the putty is dropped onto the block when it is at one end of its path.

IDENTIFY and SET UP The problem involves the motion at a given position, not a given time, so we can use energy methods. **Figure 14.16** shows our sketches. Before the putty falls, the total mechanical energy of the block–spring system is constant. In part (a), the putty–block collision is completely inelastic: The horizontal component of momentum is conserved, kinetic energy decreases, and the amount of mass that’s oscillating increases. After the collision, the total mechanical energy remains constant at its new value. In part (b) the oscillating mass also increases, but the block isn’t moving when the putty is added; there is effectively no collision at all, and no mechanical energy is lost. We find the amplitude A_2 after each collision from the final energy of the system by using Eq. (14.21) and conservation of momentum. The period T_2 after the collision is the same in both parts (a) and (b) because the final mass is the same; we find it by using Eq. (14.12).

EXECUTE (a) Before the collision the total mechanical energy of the block and spring is $E_1 = \frac{1}{2}kA_1^2$. The block is at $x = 0$, so $U = 0$ and the energy is purely kinetic (Fig. 14.16a). If we let v_1 be the speed of the block at this point, then $E_1 = \frac{1}{2}kA_1^2 = \frac{1}{2}Mv_1^2$ and

$$v_1 = \sqrt{\frac{k}{M}}A_1$$

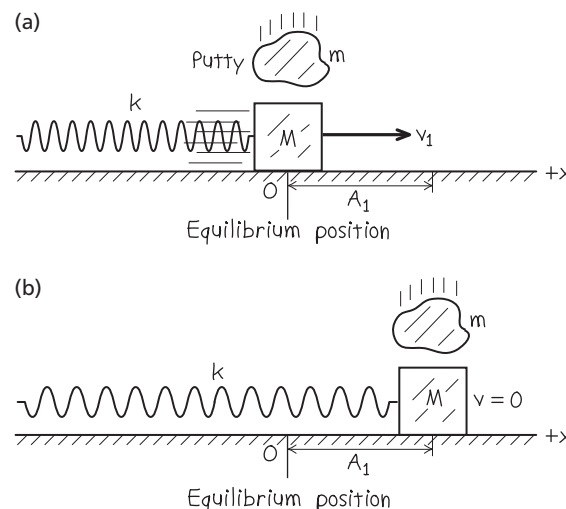
During the collision the x -component of momentum of the block–putty system is conserved. (Why?) Just before the collision this component is the sum of Mv_1 (for the block) and zero (for the putty). Just after the collision the block and putty move together with speed v_2 , so their combined x -component of momentum is $(M + m)v_2$. From conservation of momentum,

$$Mv_1 + 0 = (M + m)v_2 \quad \text{so} \quad v_2 = \frac{M}{M + m}v_1$$

The collision lasts a very short time, so the block and putty are still at the equilibrium position just after the collision. The energy is still purely kinetic but is *less* than before the collision:

$$\begin{aligned} E_2 &= \frac{1}{2}(M + m)v_2^2 = \frac{1}{2}\frac{M^2}{M + m}v_1^2 \\ &= \frac{M}{M + m}\left(\frac{1}{2}Mv_1^2\right) = \left(\frac{M}{M + m}\right)E_1 \end{aligned}$$

Figure 14.16 Our sketches for this problem.



Since $E_2 = \frac{1}{2}kA_2^2$, where A_2 is the amplitude after the collision,

$$\frac{1}{2}kA_2^2 = \left(\frac{M}{M + m}\right)\frac{1}{2}kA_1^2$$

$$A_2 = A_1\sqrt{\frac{M}{M + m}}$$

From Eq. (14.12), the period of oscillation after the collision is

$$T_2 = 2\pi\sqrt{\frac{M + m}{k}}$$

(b) When the putty falls, the block is instantaneously at rest (Fig. 14.16b). The x -component of momentum is zero both before and after the collision. The block and putty have zero kinetic energy just before and just after the collision. The energy is all potential energy stored in the spring, so adding the putty has *no effect* on the total mechanical energy. That is, $E_2 = E_1 = \frac{1}{2}kA_1^2$, and the amplitude is unchanged: $A_2 = A_1$. The period is again $T_2 = 2\pi\sqrt{(M + m)/k}$.

EVALUATE Energy is lost in part (a) because the putty slides against the moving block during the collision, and energy is dissipated by kinetic friction. No energy is lost in part (b) because there is no sliding during the collision.

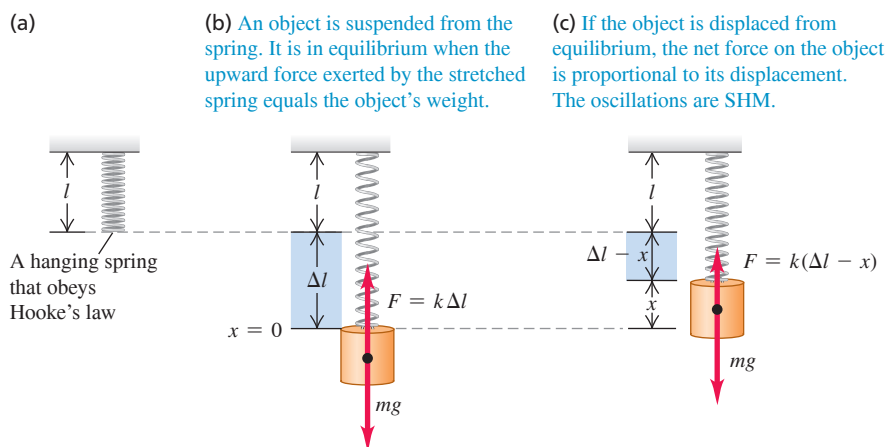
KEYCONCEPT You can use conservation of total mechanical energy in simple harmonic motion to help solve problems that involve motion at a given position, not at a given time.

TEST YOUR UNDERSTANDING OF SECTION 14.3 (a) To double the total energy for a mass-spring system oscillating in SHM, by what factor must the amplitude increase? (i) 4; (ii) 2; (iii) $\sqrt{2} = 1.414$; (iv) $\sqrt[4]{2} = 1.189$. (b) By what factor will the frequency change due to this amplitude increase? (i) 4; (ii) 2; (iii) $\sqrt{2} = 1.414$; (iv) $\sqrt[4]{2} = 1.189$; (v) it does not change.

ANSWER

(a) (i) 4; (ii) 2; (iii) $\sqrt{2} = 1.414$; (iv) $\sqrt[4]{2} = 1.189$; (v) it does not change. (b) (i) 4; (ii) 2; (iii) $\sqrt{2} = 1.414$; (iv) $\sqrt[4]{2} = 1.189$; (v) it does not change.

Figure 14.17 An object attached to a hanging spring.



14.4 APPLICATIONS OF SIMPLE HARMONIC MOTION

So far, we've looked at a grand total of *one* situation in which simple harmonic motion (SHM) occurs: an object attached to an ideal horizontal spring. But SHM can occur in any system in which there is a restoring force that is directly proportional to the displacement from equilibrium, as given by Eq. (14.3), $F_x = -kx$. The restoring force originates in different ways in different situations, so we must find the force constant k for each case by examining the net force on the system. Once this is done, it's straightforward to find the angular frequency ω , frequency f , and period T ; we just substitute the value of k into Eqs. (14.10), (14.11), and (14.12), respectively. Let's use these ideas to examine several examples of simple harmonic motion.

Vertical SHM

Suppose we hang a spring with force constant k (**Fig. 14.17a**) and suspend from it an object with mass m . Oscillations will now be vertical; will they still be SHM? In **Fig. 14.17b** the object hangs at rest, in equilibrium. In this position the spring is stretched an amount Δl just great enough that the spring's upward vertical force $k\Delta l$ on the object balances its weight mg :

$$k\Delta l = mg$$

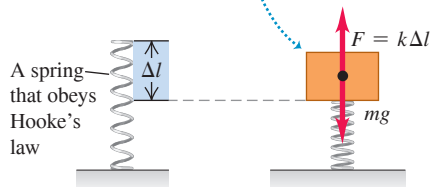
Take $x = 0$ to be this equilibrium position and take the positive x -direction to be upward. When the object is a distance x *above* its equilibrium position (**Fig. 14.17c**), the extension of the spring is $\Delta l - x$. The upward force it exerts on the object is then $k(\Delta l - x)$, and the net x -component of force on the object is

$$F_{\text{net}} = k(\Delta l - x) + (-mg) = -kx$$

that is, a net downward force of magnitude kx . Similarly, when the object is *below* the equilibrium position, there is a net upward force with magnitude kx . In either case there is a restoring force with magnitude kx . If the object is set in vertical motion, it oscillates in SHM with the same angular frequency as though it were horizontal, $\omega = \sqrt{k/m}$. So vertical SHM doesn't differ in any essential way from horizontal SHM. The only real change is that the equilibrium position $x = 0$ no longer corresponds to the point at which the spring is unstretched. The same ideas hold if an object with weight mg is placed atop a compressible spring (**Fig. 14.18**) and compresses it a distance Δl .

Figure 14.18 If the weight mg compresses the spring a distance Δl , the force constant is $k = mg/\Delta l$ and the angular frequency for vertical SHM is $\omega = \sqrt{k/m}$ —the same as if the object were suspended from the spring (see **Fig. 14.17**).

An object is placed atop the spring. It is in equilibrium when the upward force exerted by the compressed spring equals the object's weight.



EXAMPLE 14.6 Vertical SHM in an old car

The shock absorbers in an old car with mass 1000 kg are completely worn out. When a 980 N person climbs slowly into the car at its center of gravity, the car sinks 2.8 cm. The car (with the person aboard) hits a bump, and the car starts oscillating up and down in SHM. Model the car and person as a single object on a single spring, and find the period and frequency of the oscillation.

IDENTIFY and SET UP The situation is like that shown in Fig. 14.18. The compression of the spring when the person's weight is added tells us the force constant, which we can use to find the period and frequency (the target variables).

EXECUTE When the force increases by 980 N, the spring compresses an additional 0.028 m, and the x -coordinate of the car changes by -0.028 m. Hence the effective force constant (including the effect of the entire suspension) is

$$k = -\frac{F_x}{x} = -\frac{980 \text{ N}}{-0.028 \text{ m}} = 3.5 \times 10^4 \text{ kg/s}^2$$

The person's mass is $w/g = (980 \text{ N})/(9.8 \text{ m/s}^2) = 100 \text{ kg}$. The *total* oscillating mass is $m = 1000 \text{ kg} + 100 \text{ kg} = 1100 \text{ kg}$. The period T is

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{1100 \text{ kg}}{3.5 \times 10^4 \text{ kg/s}^2}} = 1.11 \text{ s}$$

The frequency is $f = 1/T = 1/(1.11 \text{ s}) = 0.90 \text{ Hz}$.

EVALUATE A persistent oscillation with a period of about 1 second makes for a very unpleasant ride. The purpose of shock absorbers is to make such oscillations die out (see Section 14.7).

KEYCONCEPT The same equations apply to both horizontal and vertical simple harmonic motion. The only effect of gravity on vertical SHM is to change the equilibrium position of the oscillating object.

Angular SHM

A mechanical watch keeps time based on the oscillations of a balance wheel (**Fig. 14.19**). The wheel has a moment of inertia I about its axis. A coil spring exerts a restoring torque τ_z that is proportional to the angular displacement θ from the equilibrium position. We write $\tau_z = -\kappa\theta$, where κ (the Greek letter kappa) is a constant called the *torsion constant*. Using the rotational analog of Newton's second law for a rigid body, $\Sigma\tau_z = I\alpha_z = I d^2\theta/dt^2$, Eq. (10.7), we find

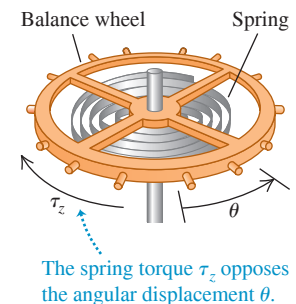
$$-\kappa\theta = I\alpha \quad \text{or} \quad \frac{d^2\theta}{dt^2} = -\frac{\kappa}{I}\theta$$

This equation is exactly the same as Eq. (14.4) for simple harmonic motion, with x replaced by θ and k/m replaced by κ/I . So we are dealing with a form of *angular* simple harmonic motion. The angular frequency ω and frequency f are given by Eqs. (14.10) and (14.11), respectively, with the same replacement:

$$\text{Angular simple harmonic motion} \quad \omega = \sqrt{\frac{\kappa}{I}} \quad \text{and} \quad f = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}} \quad (14.24)$$

Angular frequency Frequency
Torsion constant divided by moment of inertia

Figure 14.19 The balance wheel of a mechanical watch. The spring exerts a restoring torque that is proportional to the angular displacement θ , so the motion is angular SHM.



The angular displacement θ as a function of time is given by

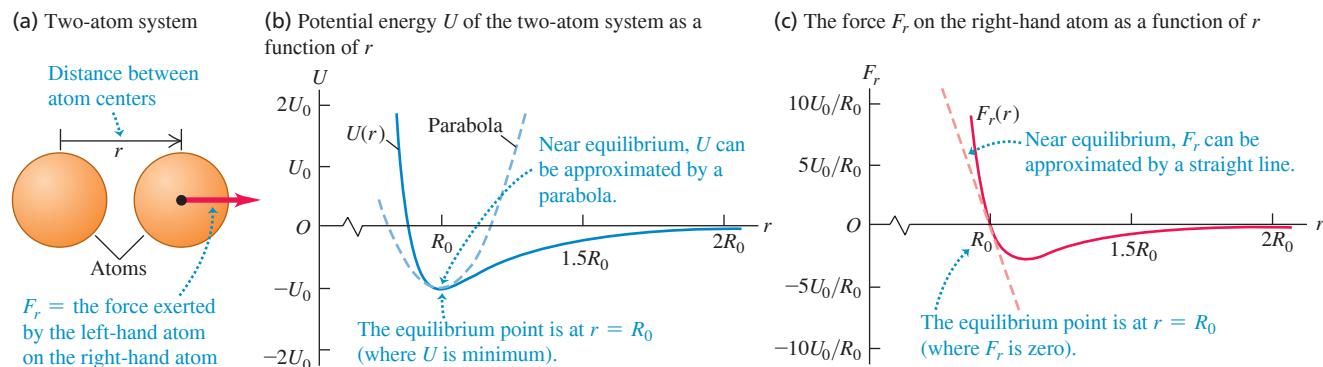
$$\theta = \Theta \cos(\omega t + \phi)$$

where Θ (the capital Greek letter theta) plays the role of an angular amplitude.

It's a good thing that the motion of a balance wheel *is* simple harmonic. If it weren't, the frequency might depend on the amplitude, and the watch would run too fast or too slow as the spring ran down.

Vibrations of Molecules

The following discussion of the vibrations of molecules uses the binomial theorem. If you aren't familiar with this theorem, you should read about it in the appropriate section of a math textbook.

Figure 14.20 (a) Two atoms with centers separated by r . (b) Potential energy U and (c) force F_r in the van der Waals interaction.

When two atoms are separated by a few atomic diameters, they can exert attractive forces on each other. But if the atoms are so close that their electron shells overlap, the atoms repel each other. Between these limits, there can be an equilibrium separation distance at which two atoms form a *molecule*. If these atoms are displaced slightly from equilibrium, they will oscillate.

Let's consider one type of interaction between atoms called the *van der Waals interaction*. Our immediate task here is to study oscillations, so we won't go into the details of how this interaction arises. Let the center of one atom be at the origin and let the center of the other atom be a distance r away (**Fig. 14.20a**); the equilibrium distance between centers is $r = R_0$. Experiment shows that the van der Waals interaction can be described by the potential-energy function

$$U = U_0 \left[\left(\frac{R_0}{r} \right)^{12} - 2 \left(\frac{R_0}{r} \right)^6 \right] \quad (14.25)$$

where U_0 is a positive constant with units of joules. When the two atoms are very far apart, $U = 0$; when they are separated by the equilibrium distance $r = R_0$, $U = -U_0$. From Section 7.4, the force on the second atom is the negative derivative of Eq. (14.25):

$$F_r = -\frac{dU}{dr} = U_0 \left[\frac{12R_0^{12}}{r^{13}} - 2\frac{6R_0^6}{r^7} \right] = 12\frac{U_0}{R_0} \left[\left(\frac{R_0}{r} \right)^{13} - \left(\frac{R_0}{r} \right)^7 \right] \quad (14.26)$$

Figures 14.20b and 14.20c plot the potential energy and force, respectively. The force is positive for $r < R_0$ and negative for $r > R_0$, so it is a *restoring* force.

Let's examine the restoring force F_r in Eq. (14.26). We let x represent the displacement from equilibrium:

$$x = r - R_0 \quad \text{so} \quad r = R_0 + x$$

In terms of x , the force F_r in Eq. (14.26) becomes

$$\begin{aligned} F_r &= 12\frac{U_0}{R_0} \left[\left(\frac{R_0}{R_0 + x} \right)^{13} - \left(\frac{R_0}{R_0 + x} \right)^7 \right] \\ &= 12\frac{U_0}{R_0} \left[\frac{1}{(1 + x/R_0)^{13}} - \frac{1}{(1 + x/R_0)^7} \right] \end{aligned} \quad (14.27)$$

This looks nothing like Hooke's law, $F_x = -kx$, so we might be tempted to conclude that molecular oscillations cannot be SHM. But let us restrict ourselves to *small-amplitude* oscillations so that the absolute value of the displacement x is small in comparison to R_0 and the absolute value of the ratio x/R_0 is much less than 1. We can then simplify Eq. (14.27) by using the *binomial theorem*:

$$(1 + u)^n = 1 + nu + \frac{n(n-1)}{2!}u^2 + \frac{n(n-1)(n-2)}{3!}u^3 + \cdots \quad (14.28)$$

If $|u|$ is much less than 1, each successive term in Eq. (14.28) is much smaller than the one it follows, and we can safely approximate $(1 + u)^n$ by just the first two terms. In Eq. (14.27), u is replaced by x/R_0 and n equals -13 or -7 , so

$$\begin{aligned}\frac{1}{(1 + x/R_0)^{13}} &= (1 + x/R_0)^{-13} \approx 1 + (-13)\frac{x}{R_0} \\ \frac{1}{(1 + x/R_0)^7} &= (1 + x/R_0)^{-7} \approx 1 + (-7)\frac{x}{R_0} \\ F_r &\approx 12\frac{U_0}{R_0}\left[\left(1 + (-13)\frac{x}{R_0}\right) - \left(1 + (-7)\frac{x}{R_0}\right)\right] = -\left(\frac{72U_0}{R_0^2}\right)x \quad (14.29)\end{aligned}$$

This is just Hooke's law, with force constant $k = 72U_0/R_0^2$. (Note that k has the correct units, J/m² or N/m.) So oscillations of molecules bound by the van der Waals interaction can be simple harmonic motion, provided that the amplitude is small in comparison to R_0 so that the approximation $|x/R_0| \ll 1$ used in the derivation of Eq. (14.29) is valid.

You can also use the binomial theorem to show that the potential energy U in Eq. (14.25) can be written as $U \approx \frac{1}{2}kx^2 + C$, where $C = -U_0$ and k is again equal to $72U_0/R_0^2$. Adding a constant to the potential-energy function has no effect on the physics, so the system of two atoms is fundamentally no different from a mass attached to a horizontal spring for which $U = \frac{1}{2}kx^2$.

EXAMPLE 14.7 Molecular vibration

Two argon atoms form the molecule Ar_2 as a result of a van der Waals interaction with $U_0 = 1.68 \times 10^{-21}$ J and $R_0 = 3.82 \times 10^{-10}$ m. Find the frequency of small oscillations of one Ar atom about its equilibrium position.

IDENTIFY and SET UP This is the situation shown in Fig. 14.20. Because the oscillations are small, we can use Eq. (14.29) to find the force constant k and Eq. (14.11) to find the frequency f of SHM.

EXECUTE From Eq. (14.29),

$$k = \frac{72U_0}{R_0^2} = \frac{72(1.68 \times 10^{-21} \text{ J})}{(3.82 \times 10^{-10} \text{ m})^2} = 0.829 \text{ J/m}^2 = 0.829 \text{ N/m}$$

(This force constant is comparable to that of a loose toy spring like a Slinky™.) From Appendix D, the average atomic mass of argon is $(39.948 \text{ u})(1.66 \times 10^{-27} \text{ kg/u}) = 6.63 \times 10^{-26} \text{ kg}$.

From Eq. (14.11), if one atom is fixed and the other oscillates,

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{0.829 \text{ N/m}}{6.63 \times 10^{-26} \text{ kg}}} = 5.63 \times 10^{11} \text{ Hz}$$

EVALUATE Our answer for f isn't quite right. If no net external force acts on the molecule, its center of mass (halfway between the atoms) doesn't accelerate, so *both* atoms must oscillate with the same amplitude in opposite directions. It turns out that we can account for this by replacing m with $m/2$ in our expression for f . This makes f larger by a factor of $\sqrt{2}$, so the correct frequency is $f = \sqrt{2}(5.63 \times 10^{11} \text{ Hz}) = 7.96 \times 10^{11} \text{ Hz}$. What's more, on the atomic scale we must use *quantum mechanics* rather than Newtonian mechanics to describe motion; happily, quantum mechanics also yields $f = 7.96 \times 10^{11} \text{ Hz}$.

KEYCONCEPT Molecular vibrations, as well as the vibrations of many other systems, are approximately simple harmonic if the oscillation amplitude is sufficiently small.

TEST YOUR UNDERSTANDING OF SECTION 14.4 A block attached to a hanging ideal spring oscillates up and down with a period of 10 s on earth. If you take the block and spring to Mars, where the acceleration due to gravity is only about 40% as large as on earth, what will be the new period of oscillation? (i) 10 s; (ii) more than 10 s; (iii) less than 10 s.

ANSWER

(i) The oscillation period of an object of mass m attached to a hanging spring of force constant k is given by $T = 2\pi\sqrt{m/k}$, the same expression as for an object attached to a horizontal spring. Neither m nor k changes when the apparatus is taken to Mars, so the period is unchanged. The only difference is that in equilibrium, the spring will stretch a shorter distance on Mars than on earth due to the weaker gravity.

Figure 14.21 The dynamics of a simple pendulum.

(a) A real pendulum



(b) An idealized simple pendulum

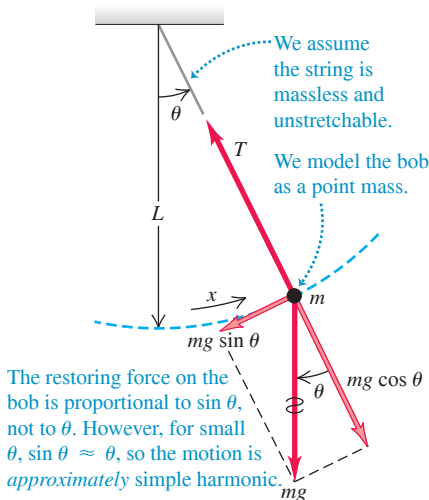
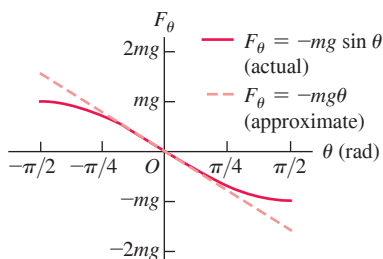


Figure 14.22 For small angular displacements θ , the restoring force $F_\theta = -mg \sin \theta$ on a simple pendulum is approximately equal to $-mg\theta$; that is, it is approximately proportional to the displacement θ . Hence for small angles the oscillations are simple harmonic.



14.5 THE SIMPLE PENDULUM

A **simple pendulum** is an idealized model consisting of a point mass suspended by a massless, unstretchable string. When the point mass is pulled to one side of its straight-down equilibrium position and released, it oscillates about the equilibrium position. Familiar situations such as a wrecking ball on a crane's cable or a person on a swing (Fig. 14.21a) can be modeled as simple pendulums.

The path of the point mass (sometimes called a *pendulum bob*) is not a straight line but the arc of a circle with radius L equal to the length of the string (Fig. 14.21b). We use as our coordinate the distance x measured along the arc. If the motion is simple harmonic, the restoring force must be directly proportional to x or (because $x = L\theta$) to θ . Is it?

Figure 14.21b shows the radial and tangential components of the forces on the mass. The restoring force F_θ is the tangential component of the net force:

$$F_\theta = -mg \sin \theta \quad (14.30)$$

Gravity provides the restoring force F_θ ; the tension T merely acts to make the point mass move in an arc. Since F_θ is proportional to $\sin \theta$, not to θ , the motion is *not* simple harmonic. However, if angle θ is *small*, $\sin \theta$ is very nearly equal to θ in radians (Fig. 14.22). (When $\theta = 0.1$ rad, about 6° , $\sin \theta = 0.0998$. That's only 0.2% different.) With this approximation, Eq. (14.30) becomes

$$F_\theta = -mg\theta = -mg \frac{x}{L} = -\frac{mg}{L}x \quad (14.31)$$

The restoring force is then proportional to the coordinate for small displacements, and the force constant is $k = mg/L$. From Eq. (14.10) the angular frequency ω of a simple pendulum with small amplitude is

$$\text{Angular frequency of simple pendulum, small amplitude} \quad \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{mg/L}{m}} = \sqrt{\frac{g}{L}} \quad (14.32)$$

Acceleration due to gravity
Pendulum length

The corresponding frequency and period relationships are

$$\text{Frequency of simple pendulum, small amplitude} \quad f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}} \quad (14.33)$$

Angular frequency
Acceleration due to gravity
Pendulum length

$$\text{Period of simple pendulum, small amplitude} \quad T = \frac{2\pi}{\omega} = \frac{1}{f} = 2\pi \sqrt{\frac{L}{g}} \quad (14.34)$$

Pendulum length
Acceleration due to gravity
Angular frequency
Frequency

These expressions don't involve the *mass* of the particle. That's because the gravitational restoring force is proportional to m , so the mass appears on *both* sides of $\Sigma \vec{F} = m\vec{a}$ and cancels out. (The same physics explains why objects of different masses fall with the same acceleration in a vacuum.) For small oscillations, the period of a pendulum for a given value of g is determined entirely by its length.

Equations (14.32) through (14.34) tell us that a long pendulum (large L) has a longer period than a shorter one. Increasing g increases the restoring force, causing the frequency to increase and the period to decrease.

The motion of a pendulum is only *approximately* simple harmonic. When the maximum angular displacement Θ (amplitude) is not small, the departures from simple harmonic motion can be substantial. In general, the period T is given by

$$T = 2\pi \sqrt{\frac{L}{g}} \left(1 + \frac{1^2}{2^2} \sin^2 \frac{\Theta}{2} + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} \sin^4 \frac{\Theta}{2} + \cdots \right) \quad (14.35)$$

We can compute T to any desired degree of precision by taking enough terms in the series. You can confirm that when $\Theta = 15^\circ$, the true period is longer than that given by the approximate Eq. (14.34) by less than 0.5%.

A pendulum is a useful timekeeper because the period is *very nearly* independent of amplitude, provided that the amplitude is small. Thus, as a pendulum clock runs down and the amplitude of the swings decreases a little, the clock still keeps very nearly correct time.

EXAMPLE 14.8 A simple pendulum

WITH VARIATION PROBLEMS

Find the period and frequency of a simple pendulum 1.000 m long at a location where $g = 9.800 \text{ m/s}^2$.

IDENTIFY and SET UP This is a simple pendulum, so we can use Eq. (14.34) to determine the pendulum's period T from its length and Eq. (14.1) to find the frequency f from T .

EXECUTE From Eqs. (14.34) and (14.1),

$$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{1.000 \text{ m}}{9.800 \text{ m/s}^2}} = 2.007 \text{ s}$$

and

$$f = \frac{1}{T} = \frac{1}{2.007 \text{ s}} = 0.4982 \text{ Hz}$$

EVALUATE The period is almost exactly 2 s. When the metric system was established, the second was *defined* as half the period of a 1 m simple pendulum. This was a poor standard, however, because the value of g varies from place to place. We discussed more modern time standards in Section 1.3.

KEYCONCEPT The period of a simple pendulum depends only on the length of the pendulum and the value of g , not on the mass of the pendulum bob.

TEST YOUR UNDERSTANDING OF SECTION 14.5 When an object oscillating on a horizontal spring passes through its equilibrium position, its acceleration is zero (see Fig. 14.2b). When the bob of an oscillating simple pendulum passes from left to right through its equilibrium position, is its acceleration (i) zero; (ii) to the left; (iii) to the right; (iv) upward; or (v) downward?

ANSWER

(iv) Just as for an object oscillating on a spring, at the equilibrium position the *speed* of a pendulum bob is instantaneously not changing (this is where the speed is maximum, so its derivative at this time is zero). But the *direction* of motion is changing because the pendulum bob follows a circular path. Hence the bob must have a component of acceleration perpendicular to the path and toward the center of the circle (see Section 3.4). To cause this acceleration at the equilibrium position when the string is vertical, the upward tension force at this position must be greater than the weight of the bob. This causes a net upward force on the bob and an upward acceleration toward the center of the circular path.

14.6 THE PHYSICAL PENDULUM

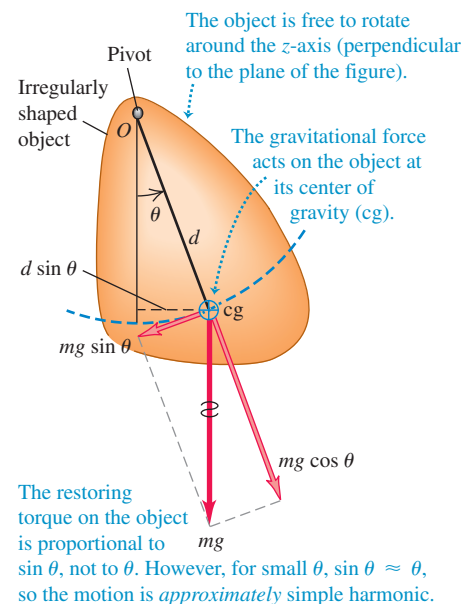
A **physical pendulum** is any *real* pendulum that uses an extended object, in contrast to the idealized *simple* pendulum with all of its mass concentrated at a point. **Figure 14.23** shows an object of irregular shape pivoted so that it can turn without friction about an axis through point O . In equilibrium the center of gravity (cg) is directly below the pivot; in the position shown, the object is displaced from equilibrium by an angle θ , which we use as a coordinate for the system. The distance from O to the center of gravity is d , the moment of inertia of the object about the axis of rotation through O is I , and the total mass is m . When the object is displaced as shown, the weight mg causes a restoring torque

$$\tau_z = -(mg)(d \sin \theta) \quad (14.36)$$

The negative sign shows that the restoring torque is clockwise when the displacement is counterclockwise, and vice versa.

When the object is released, it oscillates about its equilibrium position. The motion is not simple harmonic because the torque τ_z is proportional to $\sin \theta$ rather than to θ itself. However, if θ is small, we can approximate $\sin \theta$ by θ in radians, just as we did in

Figure 14.23 Dynamics of a physical pendulum.



analyzing the simple pendulum. Then the motion is *approximately* simple harmonic. With this approximation,

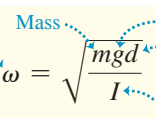
$$\tau_z = -(mgd)\theta$$

From Section 10.2, the equation of motion is $\Sigma \tau_z = I\alpha_z$, so

$$-(mgd)\theta = I\alpha_z = I\frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} = -\frac{mgd}{I}\theta \quad (14.37)$$

Comparing this with Eq. (14.4), we see that the role of (k/m) for the spring-mass system is played here by the quantity (mgd/I) . Thus the angular frequency is



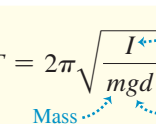
Angular frequency of physical pendulum, small amplitude

$$\omega = \sqrt{\frac{mgd}{I}}$$

Mass, Acceleration due to gravity, Distance from rotation axis to center of gravity, Moment of inertia

(14.38)

The frequency f is $1/2\pi$ times this, and the period T is $1/f$:



Period of physical pendulum, small amplitude

$$T = 2\pi\sqrt{\frac{I}{mgd}}$$

Moment of inertia, Distance from rotation axis to center of gravity, Mass, Acceleration due to gravity

(14.39)

Equation (14.39) is the basis of a common method for experimentally determining the moment of inertia of an object with a complicated shape. First locate the center of gravity by balancing the object. Then suspend the object so that it is free to oscillate about an axis, and measure the period T of small-amplitude oscillations. Finally, use Eq. (14.39) to calculate the moment of inertia I of the object about this axis from T , the object's mass m , and the distance d from the axis to the center of gravity. Biomechanics researchers use this method to find the moments of inertia of an animal's limbs. This information is important for analyzing how an animal walks, as we'll see in the second of the two following examples.

EXAMPLE 14.9 Physical pendulum versus simple pendulum

WITH VARIATION PROBLEMS

If the object in Fig. 14.23 is a uniform rod with length L , pivoted at one end, what is the period of its motion as a pendulum?

IDENTIFY and SET UP Our target variable is the oscillation period T of a rod that acts as a physical pendulum. We find the rod's moment of inertia in Table 9.2, and then determine T from Eq. (14.39).

EXECUTE The moment of inertia of a uniform rod about an axis through one end is $I = \frac{1}{3}ML^2$. The distance from the pivot to the rod's center of gravity is $d = L/2$. Then from Eq. (14.39),

$$T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{\frac{1}{3}ML^2}{MgL/2}} = 2\pi\sqrt{\frac{2L}{3g}}$$

EVALUATE If the rod is a meter stick ($L = 1.00$ m) and $g = 9.80$ m/s², then

$$T = 2\pi\sqrt{\frac{2(1.00 \text{ m})}{3(9.80 \text{ m/s}^2)}} = 1.64 \text{ s}$$

The period is smaller by a factor of $\sqrt{\frac{2}{3}} = 0.816$ than that of a simple pendulum of the same length (see Example 14.8). The rod's moment of inertia around one end, $I = \frac{1}{3}ML^2$, is one-third that of the simple pendulum, and the rod's cg is half as far from the pivot as that of the simple pendulum. You can show that, taken together in Eq. (14.39), these two differences account for the factor $\sqrt{\frac{2}{3}}$ by which the periods differ.

KEYCONCEPT The period of a physical pendulum depends on the value of g , the distance from the pivot to the physical pendulum's center of gravity, and how the mass is distributed within the physical pendulum.

EXAMPLE 14.10 *Tyrannosaurus rex* and the physical pendulum

All walking animals, including humans, have a natural walking pace—a number of steps per minute that is more comfortable than a faster or slower pace. Suppose that this pace corresponds to the oscillation of the leg as a physical pendulum. (a) How does this pace depend on the length L of the leg from hip to foot? Treat the leg as a uniform rod pivoted at the hip joint. (b) Fossil evidence shows that *T. rex*, a two-legged dinosaur that lived about 65 million years ago, had a leg length $L = 3.1$ m and a stride length $S = 4.0$ m (the distance from one footprint to the next print of the same foot; see **Fig. 14.24**). Estimate the walking speed of *T. rex*.

IDENTIFY and SET UP Our target variables are (a) the relationship between walking pace and leg length L and (b) the walking speed of *T. rex*. We treat the leg as a physical pendulum, with a period of oscillation as found in Example 14.9. We can find the walking speed from the period and the stride length.

EXECUTE (a) From Example 14.9 the period of oscillation of the leg is $T = 2\pi\sqrt{2L/3g}$, which is proportional to $\sqrt{L}$. Each step takes half of one period, so the walking pace (in steps per second) equals twice the oscillation frequency $f = 1/T$ and is proportional to $1/\sqrt{L}$. The longer the leg, the slower the pace.

(b) In our model, *T. rex* traveled one stride length S in a time

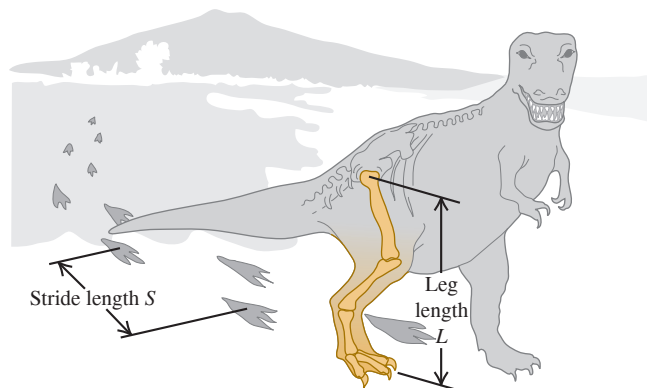
$$T = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{2(3.1\text{ m})}{3(9.8\text{ m/s}^2)}} = 2.9\text{ s}$$

so its walking speed was

$$v = \frac{S}{T} = \frac{4.0\text{ m}}{2.9\text{ s}} = 1.4\text{ m/s} = 5.0\text{ km/h} = 3.1\text{ mi/h}$$

This is roughly the walking speed of an adult human.

Figure 14.24 The walking speed of *Tyrannosaurus rex* can be estimated from leg length L and stride length S .



EVALUATE A uniform rod isn't a very good model for a leg. The legs of many animals, including both *T. rex* and humans, are tapered; there is more mass between hip and knee than between knee and foot. The center of mass is therefore less than $L/2$ from the hip; a reasonable guess would be about $L/4$. The moment of inertia is therefore considerably less than $ML^2/3$ —say, $ML^2/15$. Use the analysis of Example 14.9 with these corrections; you'll get a shorter oscillation period and an even greater walking speed for *T. rex*.

KEYCONCEPT You can apply the idea of a physical pendulum to many oscillating systems, including the swinging limbs of humans and animals.

TEST YOUR UNDERSTANDING OF SECTION 14.6 The center of gravity of a simple pendulum of mass m and length L is located at the pendulum bob, a distance L from the pivot point. The center of gravity of a uniform rod of the same mass m and length $2L$ pivoted at one end is also a distance L from the pivot point. Compared to the period of the simple pendulum, is the period of this uniform rod (i) longer; (ii) shorter; or (iii) the same?

ANSWER

(i) The period of a physical pendulum is given by Eq. (14.39), $T = 2\pi\sqrt{I/mgd}$. The distance $d = L$ from the pivot to the center of gravity is the same for both the rod and the simple pendulum, as is the mass m . Thus for any displacement angle θ the same restoring torque acts on both the rod and the simple pendulum. However, the rod has a greater moment of inertia: $I_{\text{rod}} = \frac{1}{3}m(2L)^2 = \frac{4}{3}mL^2$ and $I_{\text{simple}} = mL^2$ (all the mass of the pendulum is a distance L from the pivot). Hence the rod has a longer period.

14.7 DAMPED OSCILLATIONS

The idealized oscillating systems we have discussed so far are frictionless. There are no nonconservative forces, the total mechanical energy is constant, and a system set into motion continues oscillating forever with no decrease in amplitude.

Real-world systems always have some dissipative forces, however, and oscillations die out with time unless we replace the dissipated mechanical energy (**Fig. 14.25**). A mechanical pendulum clock continues to run because potential energy stored in the spring or a hanging weight system replaces the mechanical energy lost due to friction in the pivot and the gears. But eventually the spring runs down or the weights reach the bottom of their travel. Then no more energy is available, and the pendulum swings decrease in amplitude and stop.

Figure 14.25 A swinging bell left to itself will eventually stop oscillating due to damping forces (air resistance and friction at the point of suspension).



The decrease in amplitude caused by dissipative forces is called **damping** (not “dampening”), and the corresponding motion is called **damped oscillation**. The simplest case is a simple harmonic oscillator with a frictional damping force that is directly proportional to the *velocity* of the oscillating object. This behavior occurs in friction involving viscous fluid flow, such as in shock absorbers or sliding between oil-lubricated surfaces. We then have an additional force on the object due to friction, $F_x = -bv_x$, where $v_x = dx/dt$ is the velocity and b is a constant that describes the strength of the damping force. The negative sign shows that the force is always opposite in direction to the velocity. The *net* force on the object is then

$$\Sigma F_x = -kx - bv_x \quad (14.40)$$

and Newton’s second law for the system is

$$-kx - bv_x = ma_x \quad \text{or} \quad -kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2} \quad (14.41)$$

Equation (14.41) is a differential equation for x ; it’s the same as Eq. (14.4), the equation for the acceleration in SHM, but with the added term $-b dx/dt$. We won’t go into how to solve this equation; we’ll just present the solution. If the damping force is relatively small, the motion is described by

$$x = Ae^{-(b/2m)t} \cos(\omega' t + \phi) \quad (14.42)$$

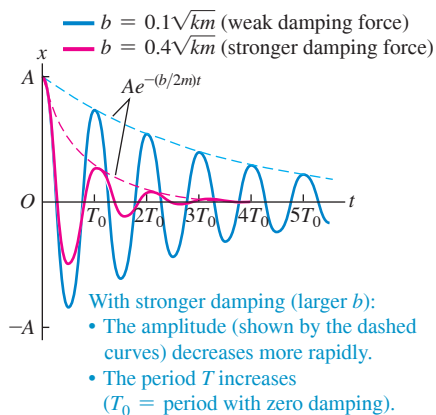
Displacement of oscillator, little damping $\rightarrow$ x $\leftarrow$ Initial amplitude $\rightarrow$ Damping constant $\rightarrow$ Mass $\rightarrow$ Time
 $\leftarrow$ Angular frequency of damped oscillations $\leftarrow$ Phase angle

The angular frequency of these damped oscillations is given by

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad (14.43)$$

Angular frequency of oscillator, little damping $\rightarrow$ ω' $\leftarrow$ Force constant of restoring force $\leftarrow$ Damping constant $\leftarrow$ Mass

Figure 14.26 Graph of displacement versus time for an oscillator with little damping [see Eq. (14.42)] and with phase angle $\phi = 0$. The curves are for two values of the damping constant b .



CAUTION When frequencies are imaginary Note that when there is overdamping and b is greater than $2\sqrt{km}$, the argument of the square root in Eq. (14.43) is negative and the angular frequency of oscillation ω is an imaginary number. This is a mathematical clue that there is *no* oscillation in this case. I

You can verify that Eq. (14.42) is a solution of Eq. (14.41) by calculating the first and second derivatives of x , substituting them into Eq. (14.41), and checking whether the left and right sides are equal.

The motion described by Eq. (14.42) differs from the undamped case in two ways. First, the amplitude $Ae^{-(b/2m)t}$ is not constant but decreases with time because of the exponential factor $e^{-(b/2m)t}$. **Figure 14.26** is a graph of Eq. (14.42) for $\phi = 0$; the larger the value of b , the more quickly the amplitude decreases.

Second, the angular frequency ω' , given by Eq. (14.43), is no longer equal to $\omega = \sqrt{k/m}$ but is somewhat smaller. It becomes zero when b becomes so large that

$$\frac{k}{m} - \frac{b^2}{4m^2} = 0 \quad \text{or} \quad b = 2\sqrt{km} \quad (14.44)$$

When Eq. (14.44) is satisfied, the condition is called **critical damping**. The system no longer oscillates but returns to its equilibrium position without oscillation when it is displaced and released.

If b is greater than $2\sqrt{km}$, the condition is called **overdamping**. Again there is no oscillation, but the system returns to equilibrium more slowly than with critical damping. For the overdamped case the solutions of Eq. (14.41) have the form

$$x = C_1 e^{-a_1 t} + C_2 e^{-a_2 t}$$

where C_1 and C_2 are constants that depend on the initial conditions and a_1 and a_2 are constants determined by m , k , and b .

When b is less than the critical value, as in Eq. (14.42), the condition is called **underdamping**. The system oscillates with steadily decreasing amplitude.

In a vibrating tuning fork or guitar string, it is usually desirable to have as little damping as possible. By contrast, damping plays a beneficial role in the oscillations of an automobile's suspension system. The shock absorbers provide a velocity-dependent damping force so that when the car goes over a bump, it doesn't continue bouncing forever (**Fig. 14.27**). For optimal passenger comfort, the system should be critically damped or slightly underdamped. Too much damping would be counterproductive; if the suspension is overdamped and the car hits a second bump just after the first one, the springs in the suspension will still be compressed somewhat from the first bump and will not be able to fully absorb the impact.

Energy in Damped Oscillations

In damped oscillations the damping force is nonconservative; the total mechanical energy of the system is not constant but decreases continuously, approaching zero after a long time. To derive an expression for the rate of change of energy, we first write an expression for the total mechanical energy E at any instant:

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2$$

To find the rate of change of this quantity, we take its time derivative:

$$\frac{dE}{dt} = mv_x \frac{dv_x}{dt} + kx \frac{dx}{dt}$$

But $dv_x/dt = a_x$ and $dx/dt = v_x$, so

$$\frac{dE}{dt} = v_x(ma_x + kx)$$

From Eq. (14.41), $ma_x + kx = -b dx/dt = -bv_x$, so

$$\frac{dE}{dt} = v_x(-bv_x) = -bv_x^2 \quad (\text{damped oscillations}) \quad (14.45)$$

The right side of Eq. (14.45) is negative whenever the oscillating object is in motion, whether the x -velocity v_x is positive or negative. This shows that as the object moves, the energy decreases, though not at a uniform rate. The term $-bv_x^2 = (-bv_x)v_x$ (force times velocity) is the rate at which the damping force does (negative) work on the system (that is, the damping power). This equals the rate of change of the total mechanical energy of the system.

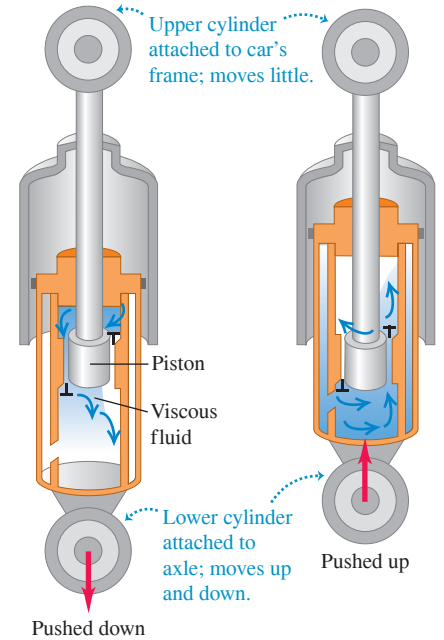
Similar behavior occurs in electric circuits containing inductance, capacitance, and resistance. There is a natural frequency of oscillation, and the resistance plays the role of the damping constant b . We'll study these circuits in detail in Chapters 30 and 31.

TEST YOUR UNDERSTANDING OF SECTION 14.7 An airplane is flying in a straight line at a constant altitude. If a wind gust strikes and raises the nose of the airplane, the nose will bob up and down until the airplane eventually returns to its original attitude. Are these oscillations (i) undamped; (ii) underdamped; (iii) critically damped; or (iv) overdamped?

ANSWER

(ii) The oscillations are underdamped with a decreasing amplitude on each cycle of oscillation, like those graphed in Fig. 14.26. If the oscillations were undamped, they would continue indefinitely with the same amplitude. If they were critically damped or overdamped, the nose would not bob up and down but would return smoothly to the original equilibrium attitude without overshooting.

Figure 14.27 An automobile shock absorber. The viscous fluid causes a damping force that depends on the relative velocity of the two ends of the unit.



14.8 FORCED OSCILLATIONS AND RESONANCE

A damped oscillator left to itself will eventually stop moving. But we can maintain a constant-amplitude oscillation by applying a force that varies with time in a periodic way. As an example, consider your cousin Throckmorton on a playground swing. You can keep him swinging with constant amplitude by giving him a push once each cycle. We call this additional force a **driving force**.

BIO APPLICATION Forced

Oscillations This lady beetle (or “ladybug,” family Coccinellidae) flies by means of a forced oscillation. Unlike the wings of birds, this insect’s wings are extensions of its exoskeleton. Muscles attached to the inside of the exoskeleton apply a periodic driving force that deforms the exoskeleton rhythmically, causing the attached wings to beat up and down. The oscillation frequency of the wings and exoskeleton is the same as the frequency of the driving force.

**Damped Oscillation with a Periodic Driving Force**

If we apply a periodic driving force with angular frequency ω_d to a damped harmonic oscillator, the motion that results is called a **forced oscillation** or a *driven oscillation*. It is different from the motion that occurs when the system is simply displaced from equilibrium and then left alone, in which case the system oscillates with a **natural angular frequency** ω' determined by m , k , and b , as in Eq. (14.43). In a forced oscillation, however, the angular frequency with which the mass oscillates is equal to the driving angular frequency ω_d . This does *not* have to be equal to the natural angular frequency ω' . If you grab the ropes of Throckmorton’s swing, you can force the swing to oscillate with any frequency you like.

Suppose we force the oscillator to vibrate with an angular frequency ω_d that is nearly *equal* to the angular frequency ω' it would have with no driving force. What happens? The oscillator is naturally disposed to oscillate at $\omega = \omega'$, so we expect the amplitude of the resulting oscillation to be larger than when the two frequencies are very different. Detailed analysis and experiment show that this is just what happens. The easiest case to analyze is a *sinusoidally* varying force—say, $F(t) = F_{\max} \cos \omega_d t$. If we vary the frequency ω_d of the driving force, the amplitude of the resulting forced oscillation varies in an interesting way (**Fig. 14.28**). When there is very little damping (small b), the amplitude goes through a sharp peak as the driving angular frequency ω_d nears the natural oscillation angular frequency ω' . When the damping is increased (larger b), the peak becomes broader and smaller in height and shifts toward lower frequencies.

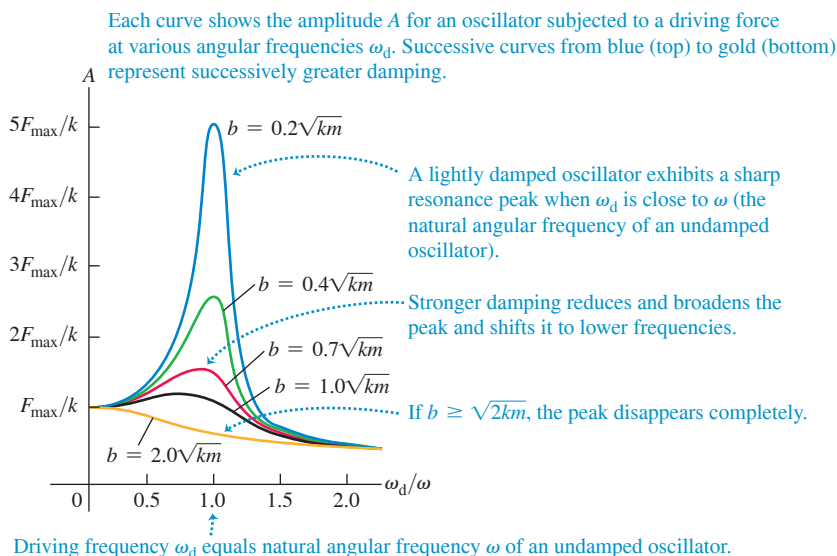
Using more differential equations than we’re ready for, we could find an expression for the amplitude A of the forced oscillation as a function of the driving angular frequency. Here is the result:

$$A = \frac{F_{\max}}{\sqrt{(k - m\omega_d^2)^2 + b^2\omega_d^2}} \quad (14.46)$$

Amplitude of a forced oscillator $\rightarrow$ A $\leftarrow$ Maximum value of driving force
Force constant of restoring force $\rightarrow$ k $\leftarrow$ Mass
Damping constant $\rightarrow$ b $\leftarrow$ Driving angular frequency

When $k - m\omega_d^2 = 0$, the first term under the radical is zero, so A has a maximum near $\omega_d = \sqrt{k/m}$. The height of the curve at this point is proportional to $1/b$; the less damping, the higher the peak. At the low-frequency extreme, when $\omega_d = 0$, we get $A = F_{\max}/k$. This corresponds to a *constant* force $F_{\max}$ and a constant displacement $A = F_{\max}/k$ from equilibrium, as we might expect.

Figure 14.28 Graph of the amplitude A of forced oscillation as a function of the angular frequency ω_d of the driving force. The horizontal axis shows the ratio of ω_d to the angular frequency $\omega = \sqrt{k/m}$ of an undamped oscillator. Each curve has a different value of the damping constant b .



Resonance and Its Consequences

The peaking of the amplitude at driving frequencies close to the natural frequency of the system is called **resonance**. Physics is full of examples of resonance; building up the oscillations of a child on a swing by pushing with a frequency equal to the swing's natural frequency is one. A vibrating rattle in a car that occurs only at a certain engine speed is another example. Inexpensive loudspeakers often have an annoying boom or buzz when a musical note coincides with the natural frequency of the speaker cone or housing. In Chapter 16 we'll study examples of resonance that involve sound. Resonance also occurs in electric circuits, as we'll see in Chapter 31; a tuned circuit in a radio receiver responds strongly to waves with frequencies near its natural frequency. This phenomenon lets us select one radio station and reject other stations.

Resonance in mechanical systems can be destructive. A company of soldiers once destroyed a bridge by marching across it in step; the frequency of their steps was close to a natural frequency of the bridge, and the resulting oscillation had large enough amplitude to tear the bridge apart. Ever since, marching soldiers have been ordered to break step before crossing a bridge. Some years ago, vibrations of the engines of a particular type of airplane had just the right frequency to resonate with the natural frequencies of its wings. Large oscillations built up, and occasionally the wings fell off.

TEST YOUR UNDERSTANDING OF SECTION 14.8 When driven at a frequency near its natural frequency, an oscillator with very little damping has a much greater response than the same oscillator with more damping. When driven at a frequency that is much higher or lower than the natural frequency, which oscillator will have the greater response: (i) the one with very little damping or (ii) the one with more damping?

ANSWER

(i) Figure 14.28 shows that the curve of amplitude versus driving frequency moves upward at all frequencies as the value of the damping constant b is decreased. Hence for fixed values of k and m , the oscillator with the least damping (smallest value of b) will have the greatest response at any driving frequency.

BIO APPLICATION Canine

Resonance Unlike humans, dogs have no sweat glands and so must pant in order to cool down. The frequency at which a dog pants is very close to the resonant frequency of its respiratory system. This causes the maximum amount of air inflow and outflow and so minimizes the effort that the dog must exert to cool itself.



CHAPTER 14 SUMMARY

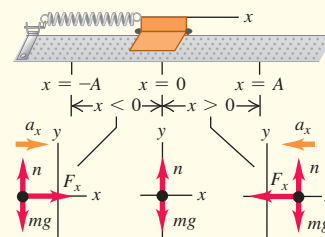
Periodic motion: Periodic motion is motion that repeats itself in a definite cycle. It occurs whenever an object has a stable equilibrium position and a restoring force that acts when the object is displaced from equilibrium. Period T is the time for one cycle. Frequency f is the number of cycles per unit time. Angular frequency ω is 2π times the frequency. (See Example 14.1.)

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

(14.1)

(14.2)



Simple harmonic motion: If the restoring force F_x in periodic motion is directly proportional to the displacement x , the motion is called simple harmonic motion (SHM). In many cases this condition is satisfied if the displacement from equilibrium is small. The angular frequency, frequency, and period in SHM do not depend on the amplitude but on only the mass m and force constant k . The displacement, velocity, and acceleration in SHM are sinusoidal functions of time; the amplitude A and phase angle ϕ of the oscillation are determined by the initial displacement and velocity of the object. (See Examples 14.2, 14.3, 14.6, and 14.7.)

$$F_x = -kx$$

(14.3)

$$a_x = \frac{F_x}{m} = -\frac{k}{m}x$$

(14.4)

$$\omega = \sqrt{\frac{k}{m}}$$

(14.10)

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

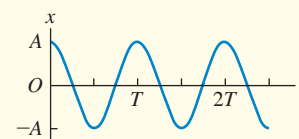
(14.11)

$$T = \frac{1}{f} = 2\pi \sqrt{\frac{m}{k}}$$

(14.12)

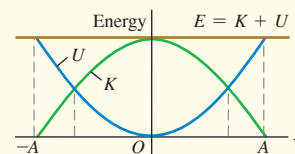
$$x = A \cos(\omega t + \phi)$$

(14.13)



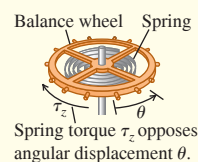
Energy in simple harmonic motion: Energy is conserved in SHM. The total energy can be expressed in terms of the force constant k and amplitude A . (See Examples 14.4 and 14.5.)

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \text{constant} \quad (14.21)$$



Angular simple harmonic motion: In angular SHM, the frequency and angular frequency are related to the moment of inertia I and the torsion constant κ .

$$\omega = \sqrt{\frac{\kappa}{I}} \quad \text{and} \quad f = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}} \quad (14.24)$$

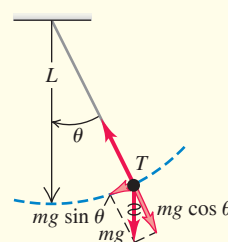


Simple pendulum: A simple pendulum consists of a point mass m at the end of a massless string of length L . Its motion is approximately simple harmonic for sufficiently small amplitude; the angular frequency, frequency, and period then depend on only g and L , not on the mass or amplitude. (See Example 14.8.)

$$\omega = \sqrt{\frac{g}{L}} \quad (14.32)$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}} \quad (14.33)$$

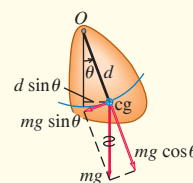
$$T = \frac{2\pi}{\omega} = \frac{1}{f} = 2\pi \sqrt{\frac{L}{g}} \quad (14.34)$$



Physical pendulum: A physical pendulum is any object suspended from an axis of rotation. The angular frequency and period for small-amplitude oscillations are independent of amplitude but depend on the mass m , distance d from the axis of rotation to the center of gravity, and moment of inertia I about the axis. (See Examples 14.9 and 14.10.)

$$\omega = \sqrt{\frac{mgd}{I}} \quad (14.38)$$

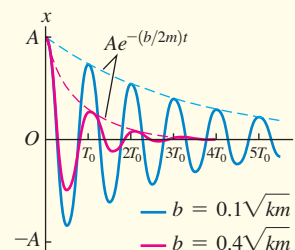
$$T = 2\pi \sqrt{\frac{I}{mgd}} \quad (14.39)$$



Damped oscillations: When a force $F_x = -bv_x$ is added to a simple harmonic oscillator, the motion is called a damped oscillation. If $b < 2\sqrt{km}$ (called underdamping), the system oscillates with a decaying amplitude and an angular frequency ω' that is lower than it would be without damping. If $b = 2\sqrt{km}$ (called critical damping) or $b > 2\sqrt{km}$ (called overdamping), when the system is displaced it returns to equilibrium without oscillating.

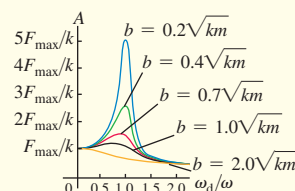
$$x = Ae^{-(b/2m)t} \cos(\omega't + \phi) \quad (14.42)$$

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \quad (14.43)$$



Forced oscillations and resonance: When a sinusoidally varying driving force is added to a damped harmonic oscillator, the resulting motion is called a forced oscillation or driven oscillation. The amplitude is a function of the driving frequency ω_d and reaches a peak at a driving frequency close to the natural frequency of the system. This behavior is called resonance.

$$A = \frac{F_{\max}}{\sqrt{(k - m\omega_d^2)^2 + b^2\omega_d^2}} \quad (14.46)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE **✓** VARIATION PROBLEMS

Be sure to review **EXAMPLES 14.2 and 14.3** (Section 14.2) before attempting these problems.

VP14.3.1 A glider of mass 0.400 kg is placed on a frictionless, horizontal air track. One end of a horizontal spring is attached to the glider, and the other end is attached to the end of the track. When released, the glider oscillates in SHM with frequency 4.15 Hz. (a) Find the period and angular frequency of the motion. (b) Find the force constant k of the spring. (c) Find the magnitude of the force that the spring exerts on the glider when the spring is stretched by 0.0200 m.

VP14.3.2 A hockey puck attached to a horizontal spring oscillates on a frictionless, horizontal surface. The spring has force constant 4.50 N/m and the oscillation period is 1.20 s. (a) What is the mass of the puck? (b) During an oscillation, the acceleration of the puck has maximum magnitude 1.20 m/s². What is the amplitude of the oscillation?

VP14.3.3 The piston of a gasoline engine oscillates in SHM with frequency 50.0 Hz. At one point in the cycle the piston is 0.0300 m from equilibrium and moving at 12.4 m/s. (a) What is the amplitude of the motion? (b) What is the maximum speed the piston attains during its oscillation?

VP14.3.4 A cat is sleeping on a platform that oscillates from side to side in SHM. The combined mass of the cat and platform is 5.00 kg, and the force constant of the horizontal spring attached to the platform that makes it oscillate is 185 N/m. Ignore friction. (a) What is the frequency of the oscillation? (b) The cat will wake up if the acceleration of the platform is greater than 1.52 m/s². What is the maximum amplitude of oscillation that will allow the cat to stay asleep?

Be sure to review **EXAMPLE 14.4** (Section 14.3) before attempting these problems.

VP14.4.1 A hockey puck oscillates on a frictionless, horizontal track while attached to a horizontal spring. The puck has mass 0.150 kg and the spring has force constant 8.00 N/m. The maximum speed of the puck during its oscillation is 0.350 m/s. (a) What is the amplitude of the oscillation? (b) What is the total mechanical energy of the oscillation?

(c) What are the potential energy and the kinetic energy of the puck when the displacement of the glider is 0.0300 m?

VP14.4.2 A block of mass 0.300 kg attached to a horizontal spring oscillates on a frictionless surface. The oscillation has amplitude 0.0440 m, and total mechanical energy $E = 6.00 \times 10^{-2}$ J. Find (a) the force constant of the spring and (b) the block's speed when the potential energy equals exactly $E/2$.

VP14.4.3 A glider attached to a horizontal spring oscillates on a horizontal air track. The total mechanical energy of the oscillation is 4.00×10^{-3} J, the amplitude of the oscillation is 0.0300 m, and the maximum speed of the glider is 0.125 m/s. (a) What are the force constant of the spring and the mass of the glider? (b) What is the maximum acceleration of the glider? (c) What is the magnitude of the glider's acceleration when the potential energy equals 3.00×10^{-3} J?

VP14.4.4 An object is undergoing SHM with amplitude A . For what values of the displacement is the kinetic energy equal to (a) $\frac{1}{3}$ of the total mechanical energy; (b) $\frac{4}{5}$ of the total mechanical energy?

Be sure to review **EXAMPLES 14.8 and 14.9** (Sections 14.5 and 14.6) before attempting these problems.

VP14.9.1 On an alien planet, a simple pendulum of length 0.500 m has oscillation frequency 0.609 Hz. Find (a) the period of the pendulum and (b) the acceleration due to gravity on this planet's surface.

VP14.9.2 What must be the length of a simple pendulum if its oscillation frequency is to be equal to that of an air-track glider of mass 0.350 kg attached to a spring of force constant 8.75 N/m?

VP14.9.3 At a bicycle repair shop, a bicycle tire of mass M and radius R is suspended from a peg on the wall. The moment of inertia of the tire around the peg is $2MR^2$. If the tire is displaced from equilibrium and starts swinging back and forth, what will be its frequency of oscillation?

VP14.9.4 A rod has length 0.900 m and mass 0.600 kg and is pivoted at one end. The rod is *not* uniform; the center of mass of the rod is not at its center but is 0.500 m from the pivot. The period of the rod's motion as a pendulum is 1.59 s. What is the moment of inertia of the rod around the pivot?

BRIDGING PROBLEM Oscillating and Rolling

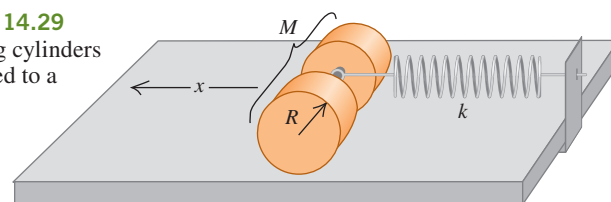
Two uniform, solid cylinders of radius R and total mass M are connected along their common axis by a short, light rod and rest on a horizontal tabletop (**Fig. 14.29**). A frictionless ring at the rod's center is attached to a spring of force constant k ; the spring's other end is fixed. The cylinders are pulled to the left a distance x , stretching the spring, then released from rest. Due to friction between the tabletop and the cylinders, the cylinders roll without slipping as they oscillate. Show that the motion of the center of mass of the cylinders is simple harmonic, and find its period.

SOLUTION GUIDE

IDENTIFY and SET UP

1. What condition must be satisfied for the motion of the center of mass of the cylinders to be simple harmonic?
2. Which equations should you use to describe (a) the translational and rotational motions of the cylinders; (b) Which equation should you use to describe the condition that the cylinders roll without slipping? (*Hint:* See Section 10.3.)
3. Sketch the situation and choose a coordinate system. List the unknown quantities and decide which is the target variable.

Figure 14.29 Rolling cylinders attached to a spring.



EXECUTE

4. Draw a free-body diagram for the cylinders when they are displaced a distance x from equilibrium.
5. Solve the equations to find an expression for the acceleration of the center of mass of the cylinders. What does this expression tell you?
6. Use your result from step 5 to find the period of oscillation of the center of mass of the cylinders.

EVALUATE

7. What would be the period of oscillation if there were no friction and the cylinders didn't roll? Is this period larger or smaller than your result from step 6? Is this reasonable?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q14.1 An object is moving with SHM of amplitude A on the end of a spring. If the amplitude is doubled, what happens to the total distance the object travels in one period? What happens to the period? What happens to the maximum speed of the object? Discuss how these answers are related.

Q14.2 Think of several examples in everyday life of motions that are, at least approximately, simple harmonic. In what respects does each differ from SHM?

Q14.3 Does a tuning fork or similar tuning instrument undergo SHM? Why is this a crucial question for musicians?

Q14.4 A box containing a pebble is attached to an ideal horizontal spring and is oscillating on a friction-free air table. When the box has reached its maximum distance from the equilibrium point, the pebble is suddenly lifted out vertically without disturbing the box. Will the following characteristics of the motion increase, decrease, or remain the same in the subsequent motion of the box? Justify each answer. (a) Frequency; (b) period; (c) amplitude; (d) the maximum kinetic energy of the box; (e) the maximum speed of the box.

Q14.5 If a uniform spring is cut in half, what is the force constant of each half? Justify your answer. How would the frequency of SHM using a half-spring differ from the frequency using the same mass and the entire spring?

Q14.6 A glider is attached to a fixed ideal spring and oscillates on a horizontal, friction-free air track. A coin rests atop the glider and oscillates with it. At what points in the motion is the friction force on the coin greatest? The least? Justify your answers.

Q14.7 Two identical gliders on an air track are connected by an ideal spring. Could such a system undergo SHM? Explain. How would the period compare with that of a single glider attached to a spring whose other end is rigidly attached to a stationary object? Explain.

Q14.8 You are captured by Martians, taken into their ship, and put to sleep. You awake some time later and find yourself locked in a small room with no windows. All the Martians have left you with is your digital watch, your school ring, and your long silver-chain necklace. Explain how you can determine whether you are still on earth or have been transported to Mars.

Q14.9 The system shown in Fig. 14.17 is mounted in an elevator. What happens to the period of the motion (does it increase, decrease, or remain the same) if the elevator (a) accelerates upward at 5.0 m/s^2 ; (b) moves upward at a steady 5.0 m/s ; (c) accelerates downward at 5.0 m/s^2 ? Justify your answers.

Q14.10 If a pendulum has a period of 2.5 s on earth, what would be its period in a space station orbiting the earth? If a mass hung from a vertical spring has a period of 5.0 s on earth, what would its period be in the space station? Justify your answers.

Q14.11 A simple pendulum is mounted in an elevator. What happens to the period of the pendulum (does it increase, decrease, or remain the same) if the elevator (a) accelerates upward at 5.0 m/s^2 ; (b) moves upward at a steady 5.0 m/s ; (c) accelerates downward at 5.0 m/s^2 ; (d) accelerates downward at 9.8 m/s^2 ? Justify your answers.

Q14.12 What should you do to the length of the string of a simple pendulum to (a) double its frequency; (b) double its period; (c) double its angular frequency?

Q14.13 If a pendulum clock is taken to a mountaintop, does it gain or lose time, assuming it is correct at a lower elevation? Explain.

Q14.14 When the amplitude of a simple pendulum increases, should its period increase or decrease? Give a qualitative argument; do not rely on Eq. (14.35). Is your argument also valid for a physical pendulum?

Q14.15 Why do short dogs (like Chihuahuas) walk with quicker strides than do tall dogs (like Great Danes)?

Q14.16 At what point in the motion of a simple pendulum is the string tension greatest? Least? In each case give the reasoning behind your answer.

Q14.17 Could a standard of time be based on the period of a certain standard pendulum? What advantages and disadvantages would such a standard have compared to the actual present-day standard discussed in Section 1.3?

Q14.18 For a simple pendulum, clearly distinguish between ω (the angular speed) and ω (the angular frequency). Which is constant and which is variable?

Q14.19 In designing structures in an earthquake-prone region, how should the natural frequencies of oscillation of a structure relate to typical earthquake frequencies? Why? Should the structure have a large or small amount of damping?

EXERCISES

Section 14.1 Describing Oscillation

14.1 • BIO (a) Music. When a person sings, his or her vocal cords vibrate in a repetitive pattern that has the same frequency as the note that is sung. If someone sings the note B flat, which has a frequency of 466 Hz , how much time does it take the person's vocal cords to vibrate through one complete cycle, and what is the angular frequency of the cords? **(b) Hearing.** When sound waves strike the eardrum, this membrane vibrates with the same frequency as the sound. The highest pitch that young humans can hear has a period of $50.0 \mu\text{s}$. What are the frequency and angular frequency of the vibrating eardrum for this sound? **(c) Vision.** When light having vibrations with angular frequency ranging from $2.7 \times 10^{15} \text{ rad/s}$ to $4.7 \times 10^{15} \text{ rad/s}$ strikes the retina of the eye, it stimulates the receptor cells there and is perceived as visible light. What are the limits of the period and frequency of this light? **(d) Ultrasound.** High-frequency sound waves (ultrasound) are used to probe the interior of the body, much as x rays do. To detect small objects such as tumors, a frequency of around 5.0 MHz is used. What are the period and angular frequency of the molecular vibrations caused by this pulse of sound?

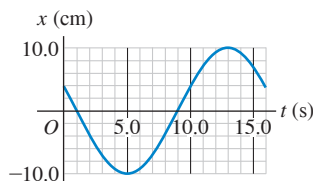
14.2 • If an object on a horizontal, frictionless surface is attached to a spring, displaced, and then released, it will oscillate. If it is displaced 0.120 m from its equilibrium position and released with zero initial speed, then after 0.800 s its displacement is found to be 0.120 m on the opposite side, and it has passed the equilibrium position once during this interval. Find (a) the amplitude; (b) the period; (c) the frequency.

14.3 • The tip of a tuning fork goes through 440 complete vibrations in 0.500 s . Find the angular frequency and the period of the motion.

14.4 • The displacement of an oscillating object as a function of time is shown in **Fig. E14.4**. What are (a) the frequency; (b) the amplitude; (c) the period; (d) the angular frequency of this motion?

14.5 •• A machine part is undergoing SHM with a frequency of 4.00 Hz and amplitude 1.80 cm. How long does it take the part to go from $x = 0$ to $x = -1.80$ cm?

Figure E14.4

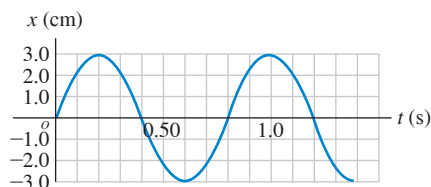


Section 14.2 Simple Harmonic Motion

14.6 • You are pushing your nephew on a playground swing. The swing seat is suspended from a horizontal bar by two light chains. Based on your experience with swings, estimate the length of each chain. Treat the motion of the child as that of a simple pendulum and assume that for safety the amplitude of the motion is kept small. You give your nephew a light push each time he reaches his closest distance from you. How much time elapses between your pushes?

14.7 • A 2.40 kg ball is attached to an unknown spring and allowed to oscillate. **Figure E14.7** shows a graph of the ball's position x as a function of time t . What are the oscillation's (a) period, (b) frequency, (c) angular frequency, and (d) amplitude? (e) What is the force constant of the spring?

Figure E14.7



14.8 •• In a physics lab, you attach a 0.200 kg air-track glider to the end of an ideal spring of negligible mass and start it oscillating. The elapsed time from when the glider first moves through the equilibrium point to the second time it moves through that point is 2.60 s. Find the spring's force constant.

14.9 • When an object of unknown mass is attached to an ideal spring with force constant 120 N/m, it is found to vibrate with a frequency of 6.00 Hz. Find (a) the period of the motion; (b) the angular frequency; (c) the mass of the object.

14.10 • When a 0.750 kg mass oscillates on an ideal spring, the frequency is 1.75 Hz. What will the frequency be if 0.220 kg are (a) added to the original mass and (b) subtracted from the original mass? Try to solve this problem *without* finding the force constant of the spring.

14.11 •• An object is undergoing SHM with period 0.900 s and amplitude 0.320 m. At $t = 0$ the object is at $x = 0.320$ m and is instantaneously at rest. Calculate the time it takes the object to go (a) from $x = 0.320$ m to $x = 0.160$ m and (b) from $x = 0.160$ m to $x = 0$.

14.12 • A small block is attached to an ideal spring and is moving in SHM on a horizontal, frictionless surface. When the block is at $x = 0.280$ m, the acceleration of the block is -5.30 m/s². What is the frequency of the motion?

14.13 • A 2.00 kg, frictionless block is attached to an ideal spring with force constant 300 N/m. At $t = 0$ the spring is neither stretched nor compressed and the block is moving in the negative direction at 12.0 m/s. Find (a) the amplitude and (b) the phase angle. (c) Write an equation for the position as a function of time.

14.14 •• Repeat Exercise 14.13, but assume that at $t = 0$ the block has velocity -4.00 m/s and displacement $+0.200$ m away from equilibrium.

14.15 •• A block of mass m is undergoing SHM on a horizontal, frictionless surface while attached to a light, horizontal spring. The spring has force constant k , and the amplitude of the SHM is A . The block has $v = 0$, and $x = +A$ at $t = 0$. It first reaches $x = 0$ when $t = T/4$, where T is the period of the motion. (a) In terms of T , what is the time t when the block first reaches $x = A/2$? (b) The block has its maximum speed when $t = T/4$. What is the value of t when the speed of the block first reaches the value $v_{\max}/2$? (c) Does $v = v_{\max}/2$ when $x = A/2$?

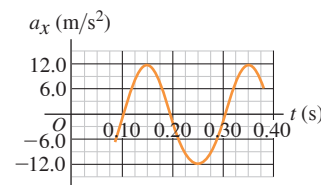
14.16 •• A small block is attached to an ideal spring and is moving in SHM on a horizontal, frictionless surface. When the amplitude of the motion is 0.090 m, it takes the block 2.70 s to travel from $x = 0.090$ m to $x = -0.090$ m. If the amplitude is doubled, to 0.180 m, how long does it take the block to travel (a) from $x = 0.180$ m to $x = -0.180$ m and (b) from $x = 0.090$ m to $x = -0.090$ m?

14.17 • BIO Weighing Astronauts. This procedure has been used to "weigh" astronauts in space: A 42.5 kg chair is attached to a spring and allowed to oscillate. When it is empty, the chair takes 1.30 s to make one complete vibration. But with an astronaut sitting in it, with her feet off the floor, the chair takes 2.54 s for one cycle. What is the mass of the astronaut?

14.18 • A 0.400 kg object undergoing SHM has $a_x = -1.80$ m/s² when $x = 0.300$ m. What is the time for one oscillation?

14.19 • On a frictionless, horizontal air track, a glider oscillates at the end of an ideal spring of force constant 2.50 N/cm. The graph in **Fig. E14.19** shows the acceleration of the glider as a function of time. Find (a) the mass of the glider; (b) the maximum displacement of the glider from the equilibrium point; (c) the maximum force the spring exerts on the glider.

Figure E14.19



14.20 • A 0.500 kg mass on a spring has velocity as a function of time given by $v_x(t) = -(3.60 \text{ cm/s}) \sin[(4.71 \text{ rad/s})t - (\pi/2)]$. What are (a) the period; (b) the amplitude; (c) the maximum acceleration of the mass; (d) the force constant of the spring?

14.21 • A 1.50 kg mass on a spring has displacement as a function of time given by

$$x(t) = (7.40 \text{ cm}) \cos[(4.16 \text{ rad/s})t - 2.42]$$

Find (a) the time for one complete vibration; (b) the force constant of the spring; (c) the maximum speed of the mass; (d) the maximum force on the mass; (e) the position, speed, and acceleration of the mass at $t = 1.00$ s; (f) the force on the mass at that time.

14.22 • BIO Weighing a Virus. In February 2004, scientists at Purdue University used a highly sensitive technique to measure the mass of a vaccinia virus (the kind used in smallpox vaccine). The procedure involved measuring the frequency of oscillation of a tiny sliver of silicon (just 30 nm long) with a laser, first without the virus and then after the virus had attached itself to the silicon. The difference in mass caused a change in the frequency. We can model such a process as a mass on a spring. (a) Show that the ratio of the frequency with the virus attached (f_{S+V}) to the frequency without the virus (f_S) is given by $f_{S+V}/f_S = 1/\sqrt{1 + (m_V/m_S)}$, where m_V is the mass of the virus and m_S is the mass of the silicon sliver. Notice that it is *not* necessary to know or measure the force constant of the spring. (b) In some data, the silicon sliver has a mass of 2.10×10^{-16} g and a frequency of 2.00×10^{15} Hz without the virus and 2.87×10^{14} Hz with the virus. What is the mass of the virus, in grams and in femtograms?

14.23 •• CALC The *jerk* is defined to be the time rate of change of the acceleration. (a) If the velocity of an object undergoing SHM is given by $v_x = -\omega A \sin(\omega t)$, what is the equation for the x -component of the jerk as a function of time? (b) What is the value of x for the object when the x -component of the jerk has its largest positive value? (c) What is x when the x -component of the jerk is most negative? (d) When is it zero? (e) If v_x equals -0.040 s² times the x -component of the jerk for all t , what is the period of the motion?

Section 14.3 Energy in Simple Harmonic Motion

14.24 •• For the oscillating object in Fig. E14.4, what are (a) its maximum speed and (b) its maximum acceleration?

14.25 • A small block is attached to an ideal spring and is moving in SHM on a horizontal, frictionless surface. The amplitude of the motion is 0.165 m. The maximum speed of the block is 3.90 m/s. What is the maximum magnitude of the acceleration of the block?

14.26 • A small block is attached to an ideal spring and is moving in SHM on a horizontal, frictionless surface. The amplitude of the motion is 0.250 m and the period is 3.20 s. What are the speed and acceleration of the block when $x = 0.160$ m?

14.27 • A 0.150 kg toy is undergoing SHM on the end of a horizontal spring with force constant $k = 300$ N/m. When the toy is 0.0120 m from its equilibrium position, it is observed to have a speed of 0.400 m/s. What are the toy's (a) total energy at any point of its motion; (b) amplitude of motion; (c) maximum speed during its motion?

14.28 •• A harmonic oscillator has angular frequency ω and amplitude A . (a) What are the magnitudes of the displacement and velocity when the elastic potential energy is equal to the kinetic energy? (Assume that $U = 0$ at equilibrium.) (b) How often does this occur in each cycle? What is the time between occurrences? (c) At an instant when the displacement is equal to $A/2$, what fraction of the total energy of the system is kinetic and what fraction is potential?

14.29 • A 0.500 kg glider, attached to the end of an ideal spring with force constant $k = 450$ N/m, undergoes SHM with an amplitude of 0.040 m. Compute (a) the maximum speed of the glider; (b) the speed of the glider when it is at $x = -0.015$ m; (c) the magnitude of the maximum acceleration of the glider; (d) the acceleration of the glider at $x = -0.015$ m; (e) the total mechanical energy of the glider at any point in its motion.

14.30 •• A block of mass m is undergoing SHM on a horizontal, frictionless surface while attached to a light, horizontal spring. The spring has force constant k , and the amplitude of the motion of the block is A . (a) The average speed is the total distance traveled by the block divided by the time it takes it to travel this distance. Calculate the average speed for one cycle of the SHM. (b) How does the average speed for one cycle compare to the maximum speed $v_{\max}$? (c) Is the average speed more or less than half the

maximum speed? Based on your answer, does the block spend more time while traveling at speeds greater than $v_{\max}/2$ or less than $v_{\max}/2$?

14.31 •• A block of mass m is undergoing SHM on a horizontal, frictionless surface while it is attached to a light, horizontal spring that has force constant k . The amplitude of the SHM of the block is A . What is the distance $|x|$ of the block from its equilibrium position when its speed v is half its maximum speed $v_{\max}$? Is this distance larger or smaller than $A/2$?

14.32 •• A block with mass $m = 0.300$ kg is attached to one end of an ideal spring and moves on a horizontal frictionless surface. The other end of the spring is attached to a wall. When the block is at $x = +0.240$ m, its acceleration is $a_x = -12.0$ m/s² and its velocity is $v_x = +4.00$ m/s. What are (a) the spring's force constant k ; (b) the amplitude of the motion; (c) the maximum speed of the block during its motion; and (d) the maximum magnitude of the block's acceleration during its motion?

14.33 •• You are watching an object that is moving in SHM. When the object is displaced 0.600 m to the right of its equilibrium position, it has a velocity of 2.20 m/s to the right and an acceleration of 8.40 m/s² to the left. How much farther from this point will the object move before it stops momentarily and then starts to move back to the left?

14.34 •• A mass is oscillating with amplitude A at the end of a spring. How far (in terms of A) is this mass from the equilibrium position of the spring when the elastic potential energy equals the kinetic energy?

14.35 • A 2.00 kg frictionless block attached to an ideal spring with force constant 315 N/m is undergoing simple harmonic motion. When the block has displacement $+0.200$ m, it is moving in the negative x -direction with a speed of 4.00 m/s. Find (a) the amplitude of the motion; (b) the block's maximum acceleration; and (c) the maximum force the spring exerts on the block.

Section 14.4 Applications of Simple Harmonic Motion

14.36 • A proud deep-sea fisherman hangs a 65.0 kg fish from an ideal spring having negligible mass. The fish stretches the spring 0.180 m. (a) Find the force constant of the spring. The fish is now pulled down 5.00 cm and released. (b) What is the period of oscillation of the fish? (c) What is the maximum speed it will reach?

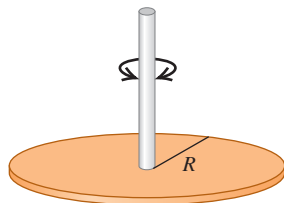
14.37 • A 175 g glider on a horizontal, frictionless air track is attached to a fixed ideal spring with force constant 155 N/m. At the instant you make measurements on the glider, it is moving at 0.815 m/s and is 3.00 cm from its equilibrium point. Use *energy conservation* to find (a) the amplitude of the motion and (b) the maximum speed of the glider. (c) What is the angular frequency of the oscillations?

14.38 •• A uniform, solid metal disk of mass 6.50 kg and diameter 24.0 cm hangs in a horizontal plane, supported at its center by a vertical metal wire. You find that it requires a horizontal force of 4.23 N tangent to the rim of the disk to turn it by 3.34°, thus twisting the wire. You now remove this force and release the disk from rest. (a) What is the torsion constant for the metal wire? (b) What are the frequency and period of the torsional oscillations of the disk? (c) Write the equation of motion for $\theta(t)$ for the disk.

14.39 • A thrill-seeking cat with mass 4.00 kg is attached by a harness to an ideal spring of negligible mass and oscillates vertically in SHM. The amplitude is 0.050 m, and at the highest point of the motion the spring has its natural unstretched length. Calculate the elastic potential energy of the spring (take it to be zero for the unstretched spring), the kinetic energy of the cat, the gravitational potential energy of the system relative to the lowest point of the motion, and the sum of these three energies when the cat is (a) at its highest point; (b) at its lowest point; (c) at its equilibrium position.

14.40 • A thin metal disk with mass 2.00×10^{-3} kg and radius 2.20 cm is attached at its center to a long fiber (**Fig. E14.40**). The disk, when twisted and released, oscillates with a period of 1.00 s. Find the torsion constant of the fiber.

Figure E14.40



14.41 •• A certain alarm clock ticks four times each second, with each tick representing half a period. The balance wheel consists of a thin rim with radius 0.55 cm, connected to the balance shaft by thin spokes of negligible mass. The total mass of the balance wheel is 0.90 g. (a) What is the moment of inertia of the balance wheel about its shaft? (b) What is the torsion constant of the coil spring (**Fig. 14.19**)?

14.42 •• You want to find the moment of inertia of a complicated machine part about an axis through its center of mass. You suspend it from a wire along this axis. The wire has a torsion constant of $0.450 \text{ N} \cdot \text{m}/\text{rad}$. You twist the part a small amount about this axis and let it go, timing 165 oscillations in 265 s. What is its moment of inertia?

Section 14.5 The Simple Pendulum

14.43 •• You pull a simple pendulum 0.240 m long to the side through an angle of 3.50° and release it. (a) How much time does it take the pendulum bob to reach its highest speed? (b) How much time does it take if the pendulum is released at an angle of 1.75° instead of 3.50° ?

14.44 •• Equation (14.35) shows that the equation $T = 2\pi\sqrt{L/g}$ for the period of a simple pendulum is an approximation that is accurate only when the angular displacement Θ of the pendulum is small. For what value of Θ is $T = 2\pi\sqrt{L/g}$ in error by 2.0%? In your calculation consider only the first correction term in Eq. (14.35).

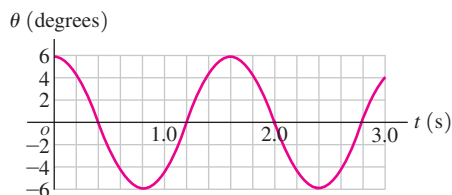
14.45 • A building in San Francisco has light fixtures consisting of small 2.35 kg bulbs with shades hanging from the ceiling at the end of light, thin cords 1.50 m long. If a minor earthquake occurs, how many swings per second will these fixtures make?

14.46 • A Pendulum on Mars. A certain simple pendulum has a period on the earth of 1.60 s. What is its period on the surface of Mars, where $g = 3.71 \text{ m/s}^2$?

14.47 • After landing on an unfamiliar planet, a space explorer constructs a simple pendulum of length 50.0 cm. She finds that the pendulum makes 100 complete swings in 136 s. What is the value of g on this planet?

14.48 •• In the laboratory, a student studies a pendulum by graphing the angle θ that the string makes with the vertical as a function of time t , obtaining the graph shown in **Fig. E14.48**. (a) What are the period, frequency, angular frequency, and amplitude of the pendulum's motion? (b) How long is the pendulum? (c) Is it possible to determine the mass of the bob?

Figure E14.48



14.49 •• A small sphere with mass m is attached to a massless rod of length L that is pivoted at the top, forming a simple pendulum. The pendulum is pulled to one side so that the rod is at an angle θ from the vertical, and released from rest. (a) In a diagram, show the pendulum just after it is released. Draw vectors representing the forces acting on the small sphere and the acceleration of the sphere. Accuracy counts! At this point, what is the linear acceleration of the sphere? (b) Repeat part (a) for the instant when the pendulum rod is at an angle $\theta/2$ from the vertical. (c) Repeat part (a) for the instant when the pendulum rod is vertical. At this point, what is the linear speed of the sphere?

Section 14.6 The Physical Pendulum

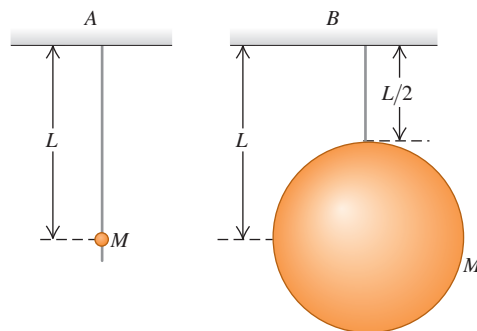
14.50 •• We want to hang a thin hoop on a horizontal nail and have the hoop make one complete small-angle oscillation each 2.0 s. What must the hoop's radius be?

14.51 • Two pendulums have the same dimensions (length L) and total mass (m). Pendulum A is a very small ball swinging at the end of a uniform massless bar. In pendulum B, half the mass is in the ball and half is in the uniform bar. Find the period of each pendulum for small oscillations. Which one takes longer for a swing?

14.52 •• A 1.80 kg monkey wrench is pivoted 0.250 m from its center of mass and allowed to swing as a physical pendulum. The period for small-angle oscillations is 0.940 s. (a) What is the moment of inertia of the wrench about an axis through the pivot? (b) If the wrench is initially displaced 0.400 rad from its equilibrium position, what is the angular speed of the wrench as it passes through the equilibrium position?

14.53 •• The two pendulums shown in **Fig. E14.53** each consist of a uniform solid ball of mass M supported by a rigid massless rod, but the ball for pendulum A is very tiny while the ball for pendulum B is much larger. Find the period of each pendulum for small displacements. Which ball takes longer to complete a swing?

Figure E14.53



14.54 •• CP A holiday ornament in the shape of a hollow sphere with mass $M = 0.015 \text{ kg}$ and radius $R = 0.050 \text{ m}$ is hung from a tree limb by a small loop of wire attached to the surface of the sphere. If the ornament is displaced a small distance and released, it swings back and forth as a physical pendulum with negligible friction. Calculate its period. (*Hint:* Use the parallel-axis theorem to find the moment of inertia of the sphere about the pivot at the tree limb.)

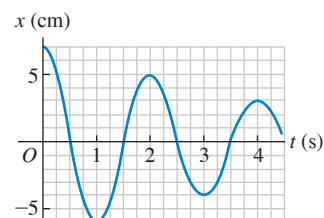
Section 14.7 Damped Oscillations

14.55 • An object is moving in damped SHM, and the damping constant can be varied. If the angular frequency of the motion is ω when the damping constant is zero, what is the angular frequency, expressed in terms of ω , when the damping constant is one-half the critical damping value?

14.56 •• A 50.0 g hard-boiled egg moves on the end of a spring with force constant $k = 25.0 \text{ N/m}$. Its initial displacement is 0.300 m. A damping force $F_x = -bv_x$ acts on the egg, and the amplitude of the motion decreases to 0.100 m in 5.00 s. Calculate the magnitude of the damping constant b .

14.57 • An unhappy 0.300 kg rodent, moving on the end of a spring with force constant $k = 2.50 \text{ N/m}$, is acted on by a damping force $F_x = -bv_x$. (a) If the constant b has the value 0.900 kg/s, what is the frequency of oscillation of the rodent? (b) For what value of the constant b will the motion be critically damped?

14.58 •• A mass is vibrating at the end of a spring of force constant 225 N/m. **Figure E14.58**



shows a graph of its position x as a function of time t . (a) At what times is the mass not moving? (b) How much energy did this system originally contain? (c) How much energy did the system lose between $t = 1.0 \text{ s}$ and $t = 4.0 \text{ s}$? Where did this energy go?

Section 14.8 Forced Oscillations and Resonance

14.59 • A sinusoidally varying driving force is applied to a damped harmonic oscillator of force constant k and mass m . If the damping constant has a value b_1 , the amplitude is A_1 when the driving angular frequency equals $\sqrt{k/m}$. In terms of A_1 , what is the amplitude for the same driving frequency and the same driving force amplitude F_{max} , if the damping constant is (a) $3b_1$ and (b) $b_1/2$?

14.60 •• Equation (14.46) and Fig. 14.28 describe a damped and driven oscillator. (a) For a damping constant $b = 0.20\sqrt{km}$, confirm that the amplitude A is $5F_{\text{max}}/k$ when $\omega_d = \omega$, where $\omega = \sqrt{k/m}$ is the natural angular frequency. (b) Repeat part (a) for $b = 0.40\sqrt{km}$, and confirm that the amplitude A is $2.5F_{\text{max}}/k$ when $\omega_d = \omega$. (c) As a measure of the width of the resonance peak, calculate A when $\omega_d = \omega/2$ for $b = 0.20\sqrt{km}$ and for $b = 0.40\sqrt{km}$. In each case, what is the ratio of the amplitude for $\omega_d = \omega$ to the amplitude for $\omega_d = \omega/2$? For which value of the damping constant does the amplitude increase by the larger factor?

PROBLEMS

14.61 •• Object A has mass m_A and is in SHM on the end of a spring with force constant k_A . Object B has mass m_B and is in SHM on the end of a spring with force constant k_B . The amplitude A_A for object A is twice the amplitude A_B for the motion of object B . Also, $m_B = 4m_A$ and $k_A = 9k_B$. (a) What is the ratio of the maximum speeds of the two objects, $v_{\text{max},A}/v_{\text{max},B}$? (b) What is the ratio of their maximum accelerations, $a_{\text{max},A}/a_{\text{max},B}$?

14.62 ••• An object is undergoing SHM with period 0.300 s and amplitude 6.00 cm. At $t = 0$ the object is instantaneously at rest at $x = 6.00 \text{ cm}$. Calculate the time it takes the object to go from $x = 6.00 \text{ cm}$ to $x = -1.50 \text{ cm}$.

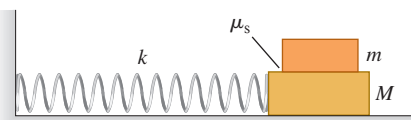
14.63 •• An object is undergoing SHM with period 1.200 s and amplitude 0.600 m. At $t = 0$ the object is at $x = 0$ and is moving in the negative x -direction. How far is the object from the equilibrium position when $t = 0.480 \text{ s}$?

14.64 • Four passengers with combined mass 250 kg compress the springs of a car with worn-out shock absorbers by 4.00 cm when they get in. Model the car and passengers as a single object on a single ideal spring. If the loaded car has a period of vibration of 1.92 s, what is the period of vibration of the empty car?

14.65 •• An object with mass m is moving in SHM. It has amplitude A_1 and total mechanical energy E_1 when the spring has force constant k_1 . You want to quadruple the total mechanical energy, so $E_2 = 4E_1$, and halve the amplitude, so $A_2 = A_1/2$, by using a different spring, one with force constant k_2 . (a) How is k_2 related to k_1 ? (b) What effect will the change in spring constant and amplitude have on the maximum speed of the moving object?

14.66 •• CP A block with mass M rests on a frictionless surface and is connected to a horizontal spring of force constant k . The other end of the spring is attached to a wall (**Fig. P14.66**). A second block with mass m rests on top of the first block. The coefficient of static friction between the blocks is μ_s . Find the *maximum* amplitude of oscillation such that the top block will not slip on the bottom block.

Figure P14.66



14.67 •• A block with mass m is undergoing SHM on a horizontal, frictionless surface while attached to a light, horizontal spring that has force constant k . You use motion sensor equipment to measure the maximum speed of the block during its oscillations. You repeat the measurement for the same spring and blocks of different masses while keeping the amplitude A at a constant value of 12.0 cm. You plot your data as v_{max}^2 versus $1/m$ and find that the data lie close to a straight line that has slope $8.62 \text{ N} \cdot \text{m}$. What is the force constant k of the spring?

14.68 •• CP Consider the system of two blocks and a spring shown in Fig. P14.66. The horizontal surface is frictionless, but there is static friction between the two blocks. The spring has force constant $k = 150 \text{ N/m}$. The masses of the two blocks are $m = 0.500 \text{ kg}$ and $M = 4.00 \text{ kg}$. You set the blocks into motion by releasing block M with the spring stretched a distance d from equilibrium. You start with small values of d , and then repeat with successively larger values. For small values of d , the blocks move together in SHM. But for larger values of d the top block slips relative to the bottom block when the bottom block is released. (a) What is the period of the motion of the two blocks when d is small enough to have no slipping? (b) The largest value d can have and there be no slipping is $d = 8.8 \text{ cm}$. What is the coefficient of static friction μ_s between the surfaces of the two blocks?

14.69 ••• A 1.50 kg, horizontal, uniform tray is attached to a vertical ideal spring of force constant 185 N/m and a 275 g metal ball is in the tray. The spring is below the tray, so it can oscillate up and down. The tray is then pushed down to point A , which is 15.0 cm below the equilibrium point, and released from rest. (a) How high above point A will the tray be when the metal ball leaves the tray? (*Hint:* This does *not* occur when the ball and tray reach their maximum speeds.) (b) How much time elapses between releasing the system at point A and the ball leaving the tray? (c) How fast is the ball moving just as it leaves the tray?

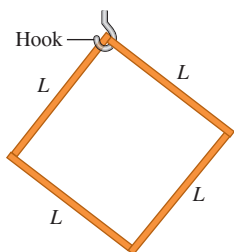
14.70 • CP A 10.0 kg mass is traveling to the right with a speed of 2.00 m/s on a smooth horizontal surface when it collides with and sticks to a second 10.0 kg mass that is initially at rest but is attached to one end of a light, horizontal spring with force constant 170.0 N/m. The other end of the spring is fixed to a wall to the right of the second mass. (a) Find the frequency, amplitude, and period of the subsequent oscillations. (b) How long does it take the system to return the first time to the position it had immediately after the collision?

14.71 ••• An apple weighs 1.00 N. When you hang it from the end of a long spring of force constant 1.50 N/m and negligible mass, it bounces up and down in SHM. If you stop the bouncing and let the apple swing from side to side through a small angle, the frequency of this simple pendulum is half the bounce frequency. (Because the angle is small, the back-and-forth swings do not cause any appreciable change in the length of the spring.) What is the unstretched length of the spring (with the apple removed)?

14.72 ••• CP SHM of a Floating Object. An object with height h , mass M , and a uniform cross-sectional area A floats upright in a liquid with density ρ . (a) Calculate the vertical distance from the surface of the liquid to the bottom of the floating object at equilibrium. (b) A downward force with magnitude F is applied to the top of the object. At the new equilibrium position, how much farther below the surface of the liquid is the bottom of the object than it was in part (a)? (Assume that some of the object remains above the surface of the liquid.) (c) Your result in part (b) shows that if the force is suddenly removed, the object will oscillate up and down in SHM. Calculate the period of this motion in terms of the density ρ of the liquid, the mass M , and the cross-sectional area A of the object. You can ignore the damping due to fluid friction (see Section 14.7).

14.73 •• CP A square object of mass m is constructed of four identical uniform thin sticks, each of length L , attached together. This object is hung on a hook at its upper corner (**Fig. P14.73**). If it is rotated slightly to the left and then released, at what frequency will it swing back and forth?

Figure P14.73



14.74 ••• An object with mass 0.200 kg is acted on by an elastic restoring force with force constant 10.0 N/m. (a) Graph elastic potential energy U as a function of displacement x over a range of x from -0.300 m to $+0.300$ m. On your graph, let 1 cm = 0.05 J vertically and 1 cm = 0.05 m horizontally. The object is set into oscillation with an initial potential energy of 0.140 J and an initial kinetic energy of 0.060 J. Answer the following questions by referring to the graph. (b) What is the amplitude of oscillation? (c) What is the potential energy when the displacement is one-half the amplitude? (d) At what displacement are the kinetic and potential energies equal? (e) What is the value of the phase angle ϕ if the initial velocity is positive and the initial displacement is negative?

14.75 • CALC A 2.00 kg bucket containing 10.0 kg of water is hanging from a vertical ideal spring of force constant 450 N/m and oscillating up and down with an amplitude of 3.00 cm. Suddenly the bucket springs a leak in the bottom such that water drops out at a steady rate of 2.00 g/s. When the bucket is half full, find (a) the period of oscillation and (b) the rate at which the period is changing with respect to time. Is the period getting longer or shorter? (c) What is the shortest period this system can have?

14.76 •• Quantum mechanics is used to describe the vibrational motion of molecules, but analysis using classical physics gives some useful insight. In a classical model the vibrational motion can be treated as SHM of the atoms connected by a spring. The two atoms in a diatomic molecule vibrate about their center of mass, but in the molecule HI,

where one atom is much more massive than the other, we can treat the hydrogen atom as oscillating in SHM while the iodine atom remains at rest. (a) A classical estimate of the vibrational frequency is $f = 7 \times 10^{13}$ Hz. The mass of a hydrogen atom differs little from the mass of a proton. If the HI molecule is modeled as two atoms connected by a spring, what is the force constant of the spring? (b) The vibrational energy of the molecule is measured to be about 5×10^{-20} J. In the classical model, what is the maximum speed of the H atom during its SHM? (c) What is the amplitude of the vibrational motion? How does your result compare to the equilibrium distance between the two atoms in the HI molecule, which is about 1.6×10^{-10} m?

14.77 •• A 5.00 kg partridge is suspended from a pear tree by an ideal spring of negligible mass. When the partridge is pulled down 0.100 m below its equilibrium position and released, it vibrates with a period of 4.20 s. (a) What is its speed as it passes through the equilibrium position? (b) What is its acceleration when it is 0.050 m above the equilibrium position? (c) When it is moving upward, how much time is required for it to move from a point 0.050 m below its equilibrium position to a point 0.050 m above it? (d) The motion of the partridge is stopped, and then it is removed from the spring. How much does the spring shorten?

14.78 •• A slender rod of length 80.0 cm and mass 0.400 kg has its center of gravity at its geometrical center. But its density is not uniform; it increases by the same amount from the center of the rod out to either end. You want to determine the moment of inertia I_{cm} of the rod for an axis perpendicular to the rod at its center, but you don't know its density as a function of distance along the rod, so you can't use an integration method to calculate I_{cm} . Therefore, you make the following measurements: You suspend the rod about an axis that is a distance d (measured in meters) above the center of the rod and measure the period T (measured in seconds) for small-amplitude oscillations about the axis. You repeat this for several values of d . When you plot your data as $T^2 - 4\pi^2 d/g$ versus $1/d$, the data lie close to a straight line that has slope $0.320 \text{ m} \cdot \text{s}^2$. What is the value of I_{cm} for the rod?

14.79 •• CP SHM of a Butcher's Scale. A spring of negligible mass and force constant $k = 400$ N/m is hung vertically, and a 0.200 kg pan is suspended from its lower end. A butcher drops a 2.2 kg steak onto the pan from a height of 0.40 m. The steak makes a totally inelastic collision with the pan and sets the system into vertical SHM. What are (a) the speed of the pan and steak immediately after the collision; (b) the amplitude of the subsequent motion; (c) the period of that motion?

14.80 •• A 40.0 N force stretches a vertical spring 0.250 m. (a) What mass must be suspended from the spring so that the system will oscillate with a period of 1.00 s? (b) If the amplitude of the motion is 0.050 m and the period is that specified in part (a), where is the object and in what direction is it moving 0.35 s after it has passed the equilibrium position, moving downward? (c) What force (magnitude and direction) does the spring exert on the object when it is 0.030 m below the equilibrium position, moving upward?

14.81 •• Don't Miss the Boat. While on a visit to Minnesota ("Land of 10,000 Lakes"), you sign up to take an excursion around one of the larger lakes. When you go to the dock where the 1500 kg boat is tied, you find that the boat is bobbing up and down in the waves, executing simple harmonic motion with amplitude 20 cm. The boat takes 3.5 s to make one complete up-and-down cycle. When the boat is at its highest point, its deck is at the same height as the stationary dock. As you watch the boat bob up and down, you (mass 60 kg) begin to feel a bit woozy, due in part to the previous night's dinner of lutefisk. As a result, you refuse to board the boat unless the level of the boat's deck is within 10 cm of the dock level. How much time do you have to board the boat comfortably during each cycle of up-and-down motion?

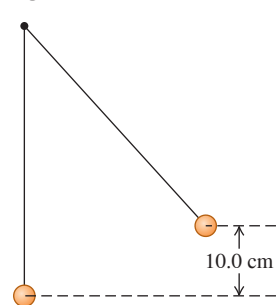
14.82 • CP An interesting, though highly impractical example of oscillation is the motion of an object dropped down a hole that extends from one side of the earth, through its center, to the other side. With the assumption (not realistic) that the earth is a sphere of uniform density, prove that the motion is simple harmonic and find the period. [Note: The gravitational force on the object as a function of the object's distance r from the center of the earth was derived in Example 13.10 (Section 13.6). The motion is simple harmonic if the acceleration a_x and the displacement from equilibrium x are related by Eq. (14.8), and the period is then $T = 2\pi/\omega$.]

14.83 •• CP A rifle bullet with mass 8.00 g and initial horizontal velocity 280 m/s strikes and embeds itself in a block with mass 0.992 kg that rests on a frictionless surface and is attached to one end of an ideal spring. The other end of the spring is attached to the wall. The impact compresses the spring a maximum distance of 15.0 cm. After the impact, the block moves in SHM. Calculate the period of this motion.

14.84 •• CP Two uniform solid spheres, each with mass $M = 0.800$ kg and radius $R = 0.0800$ m, are connected by a short, light rod that is along a diameter of each sphere and are at rest on a horizontal tabletop. A spring with force constant $k = 160$ N/m has one end attached to the wall and the other end attached to a frictionless ring that passes over the rod at the center of mass of the spheres, which is midway between the centers of the two spheres. The spheres are each pulled the same distance from the wall, stretching the spring, and released. There is sufficient friction between the tabletop and the spheres for the spheres to roll without slipping as they move back and forth on the end of the spring. Show that the motion of the center of mass of the spheres is simple harmonic and calculate the period.

14.85 • CP In Fig. P14.85 the upper ball is released from rest, collides with the stationary lower ball, and sticks to it. The strings are both 50.0 cm long. The upper ball has mass 2.00 kg, and it is initially 10.0 cm higher than the lower ball, which has mass 3.00 kg. Find the frequency and maximum angular displacement of the motion after the collision.

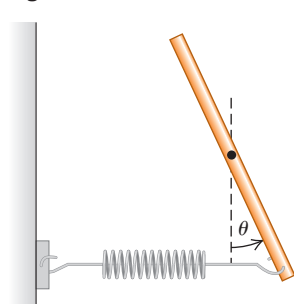
Figure P14.85



14.86 •• The Silently Ringing Bell. A large, 34.0 kg bell is hung from a wooden beam so it can swing back and forth with negligible friction. The bell's center of mass is 0.60 m below the pivot. The bell's moment of inertia about an axis at the pivot is $18.0 \text{ kg} \cdot \text{m}^2$. The clapper is a small, 1.8 kg mass attached to one end of a slender rod of length L and negligible mass. The other end of the rod is attached to the inside of the bell; the rod can swing freely about the same axis as the bell. What should be the length L of the clapper rod for the bell to ring silently—that is, for the period of oscillation for the bell to equal that of the clapper?

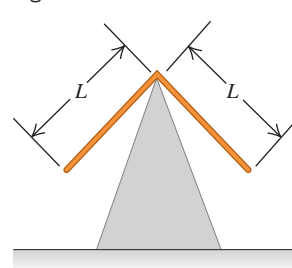
14.87 •• CALC A slender, uniform, metal rod with mass M is pivoted without friction about an axis through its midpoint and perpendicular to the rod. A horizontal spring with force constant k is attached to the lower end of the rod, with the other end of the spring attached to a rigid support. If the rod is displaced by a small angle θ from the vertical (Fig. P14.87) and released, show that it moves in angular SHM and calculate the period. (Hint: Assume that the angle θ is small enough for the approximations $\sin \theta \approx \theta$ and $\cos \theta \approx 1$ to be valid. The motion is simple harmonic if $d^2\theta/dt^2 = -\omega^2\theta$, and the period is then $T = 2\pi/\omega$.)

Figure P14.87



14.88 ••• Two identical thin rods, each with mass m and length L , are joined at right angles to form an L-shaped object. This object is balanced on top of a sharp edge (Fig. P14.88). If the L-shaped object is deflected slightly, it oscillates. Find the frequency of oscillation.

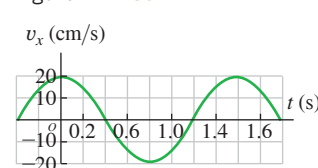
Figure P14.88



14.89 •• DATA A mass m is attached to a spring of force constant 75 N/m and allowed to oscillate.

Figure P14.89 shows a graph of its velocity component v_x as a function of time t . Find (a) the period, (b) the frequency, and (c) the angular frequency of this motion. (d) What is the amplitude (in cm), and at what times does the mass reach this position? (e) Find the maximum acceleration magnitude of the mass and the times at which it occurs. (f) What is the value of m ?

Figure P14.89



14.90 •• DATA You hang various masses m from the end of a vertical, 0.250 kg spring that obeys Hooke's law and is tapered, which means the diameter changes along the length of the spring. Since the mass of the spring is not negligible, you must replace m in the equation $T = 2\pi\sqrt{m/k}$ with $m + m_{\text{eff}}$, where m_{eff} is the effective mass of the oscillating spring. (See Challenge Problem 14.93.) You vary the mass m and measure the time for 10 complete oscillations, obtaining these data:

m (kg)	0.100	0.200	0.300	0.400	0.500
Time (s)	8.7	10.5	12.2	13.9	15.1

(a) Graph the square of the period T versus the mass suspended from the spring, and find the straight line of best fit. (b) From the slope of that line, determine the force constant of the spring. (c) From the vertical intercept of the line, determine the spring's effective mass. (d) What fraction is m_{eff} of the spring's mass? (e) If a 0.450 kg mass oscillates on the end of the spring, find its period, frequency, and angular frequency.

14.91 ••• DATA Experimenting with pendulums, you attach a light string to the ceiling and attach a small metal sphere to the lower end of the string. When you displace the sphere 2.00 m to the left, it nearly touches a vertical wall; with the string taut, you release the sphere from rest. The sphere swings back and forth as a simple pendulum, and you measure its period T . You repeat this act for strings of various lengths L , each time starting the motion with the sphere displaced 2.00 m to the left of the vertical position of the string. In each case the sphere's radius is very small compared with L . Your results are given in the table:

L (m)	12.00	10.00	8.00	6.00	5.00	4.00	3.00	2.50	2.30
T (s)	6.96	6.36	5.70	4.95	4.54	4.08	3.60	3.35	3.27

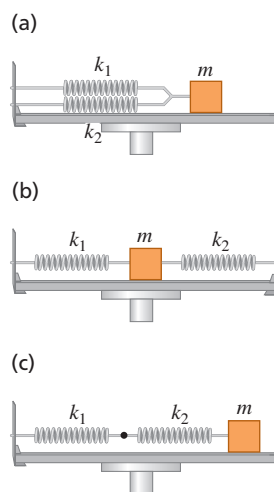
(a) For the five largest values of L , graph T^2 versus L . Explain why the data points fall close to a straight line. Does the slope of this line have the value you expected? (b) Add the remaining data to your graph. Explain why the data start to deviate from the straight-line fit as L decreases. To see this effect more clearly, plot T/T_0 versus L , where $T_0 = 2\pi\sqrt{L/g}$ and $g = 9.80 \text{ m/s}^2$. (c) Use your graph of T/T_0 versus L to estimate the angular amplitude of the pendulum (in degrees) for which the equation $T = 2\pi\sqrt{L/g}$ is in error by 5%.

CHALLENGE PROBLEMS

14.92 ••• The Effective Force Constant of Two Springs. Two springs with the same unstretched length but different force constants k_1 and k_2 are attached to a block with mass m on a level, frictionless surface. Calculate the effective force constant k_{eff} in each of the three cases (a), (b), and (c) depicted in **Fig. P14.92**. (The effective force constant is defined by $\Sigma F_x = -k_{\text{eff}}x$.) (d) An object with mass m , suspended from a uniform spring with a force constant k , vibrates with a frequency f_1 . When the spring is cut in half and the same object is suspended from one of the halves, the frequency is f_2 . What is the ratio f_1/f_2 ?

14.93 ••• CALC A Spring with Mass. The preceding problems in this chapter have assumed that the springs had negligible mass. But of course no spring is completely massless. To find the effect of the spring's mass, consider a spring with mass M , equilibrium length L_0 , and spring constant k . When stretched or compressed to a length L , the potential energy is $\frac{1}{2}kx^2$, where $x = L - L_0$. (a) Consider a spring, as described above, that has one end fixed and the other end moving with speed v . Assume that the speed of points along the length of the spring varies linearly with distance l from the fixed end. Assume also that the mass M of the spring is distributed uniformly along the length of the spring. Calculate the kinetic energy of the spring in terms of M and v . (*Hint:* Divide the spring into pieces of length dl ; find the speed of each piece in terms of l , v , and L ; find the mass of each piece in terms of dl , M , and L ; and integrate from 0 to L . The result is *not* $\frac{1}{2}Mv^2$, since not all of the spring moves with the same speed.) (b) Take the time derivative of the conservation of energy equation, Eq. (14.21), for a mass m moving on the end of a *massless* spring. By comparing your results to Eq. (14.8), which defines ω ,

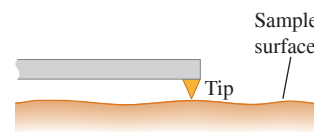
Figure P14.92



show that the angular frequency of oscillation is $\omega = \sqrt{k/m}$. (c) Apply the procedure of part (b) to obtain the angular frequency of oscillation ω of the spring considered in part (a). If the *effective mass* M' of the spring is defined by $\omega = \sqrt{k/M'}$, what is M' in terms of M ?

MCAT-STYLE PASSAGE PROBLEMS

BIO “Seeing” Surfaces at the Nanoscale. One technique for making images of surfaces at the nanometer scale, including membranes and biomolecules, is dynamic atomic force microscopy. In this technique, a small tip is attached to a cantilever, which is a flexible, rectangular slab supported at one end, like a diving board. The cantilever vibrates, so the tip moves up and down in simple harmonic motion. In one operating mode, the resonant frequency for a cantilever with force constant $k = 1000 \text{ N/m}$ is 100 kHz. As the oscillating tip is brought within a few nanometers of the surface of a sample (as shown in the figure), it experiences an attractive force from the surface. For an oscillation with a small amplitude (typically, 0.050 nm), the force F that the sample surface exerts on the tip varies linearly with the displacement x of the tip, $|F| = k_{\text{surf}}x$, where k_{surf} is the effective force constant for this force. The net force on the tip is therefore $(k + k_{\text{surf}})x$, and the frequency of the oscillation changes slightly due to the interaction with the surface. Measurements of the frequency as the tip moves over different parts of the sample's surface can provide information about the sample.



14.94 If we model the vibrating system as a mass on a spring, what is the mass necessary to achieve the desired resonant frequency when the tip is not interacting with the surface? (a) 25 ng; (b) 100 ng; (c) 2.5 μg ; (d) 100 μg .

14.95 In the model of Problem 14.94, what is the total mechanical energy of the vibration when the tip is not interacting with the surface? (a) $1.2 \times 10^{-18} \text{ J}$; (b) $1.2 \times 10^{-16} \text{ J}$; (c) $1.2 \times 10^{-9} \text{ J}$; (d) $5.0 \times 10^{-8} \text{ J}$.

14.96 By what percentage does the frequency of oscillation change if $k_{\text{surf}} = 5 \text{ N/m}$? (a) 0.1%; (b) 0.2%; (c) 0.5%; (d) 1.0%.

ANSWERS

Chapter Opening Question ?

(i) The back-and-forth motion of a leg during walking is like a physical pendulum, for which the oscillation period is $T = 2\pi\sqrt{I/mgd}$ [see Eq. (14.39)]. In this expression I is the moment of inertia of the pendulum, m is its mass, and d is the distance from the rotation axis to the pendulum center of mass. I is proportional to m , so the mass cancels out of this expression for T . Hence only the dimensions of the leg matter. (See Examples 14.9 and 14.10.)

Key Example VARIATION Problems

VP14.3.1 (a) $T = 0.241 \text{ s}$, $\omega = 26.1 \text{ rad/s}$ (b) 272 N/m (c) 5.44 N

VP14.3.2 (a) 0.164 kg (b) 0.0438 m

VP14.3.3 (a) 0.0496 m (b) 15.6 m/s

VP14.3.4 (a) 0.968 Hz (b) $4.11 \times 10^{-2} \text{ m}$

VP14.4.1 (a) 0.0479 m (b) $9.19 \times 10^{-3} \text{ J}$ (c) $U = 3.60 \times 10^{-3} \text{ J}$, $K = 5.59 \times 10^{-3} \text{ J}$

VP14.4.2 (a) 62.0 N/m (b) 0.447 m/s

VP14.4.3 (a) $k = 8.89 \text{ N/m}$, $m = 0.512 \text{ kg}$ (b) 0.521 m/s^2 (c) 0.451 m/s^2

VP14.4.4 (a) $x = \pm A\sqrt{2/3}$ (b) $x = \pm A\sqrt{1/5}$

VP14.9.1 (a) 1.64 s (b) 7.32 m/s^2

VP14.9.2 0.392 m

VP14.9.3 $\frac{1}{2\pi}\sqrt{\frac{g}{2R}}$

VP14.9.4 0.188 $\text{kg} \cdot \text{m}^2$

BRIDGING PROBLEM

$$T = 2\pi\sqrt{3M/2k}$$